Qualification Pack

Automobile Jr. Technician (Service and Maintenance)

Semester 1: Two-Wheeler Technology/ Semester 1: Engineering Drawing/ Semester 2: Electric Two or Three Wheeler Technology/ Semester 2: Four-Wheeler Technology

QP Code: ASC/Q1443

Version: 1.0

NSQF Level: 3.5

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Contents

ASC/Q1443: Automobile Jr. Technician (Service and Maintenance)

Brief Job Description

The individual in this job is responsible for Conduct thorough inspections of automobiles to identify issues, diagnose problems, and determine necessary repairs or maintenance tasks & Perform routine maintenance tasks such as oil changes, filter replacements, fluid checks and top-ups, tire rotations, and brake inspections to ensure vehicles operate efficiently and safely.

Personal Attributes

The person should be patient, organized, team-oriented and have the ability to work for long hours in adverse conditions. They should be keen observers and have an eye for detail and quality

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. ASC/N6314: Metrology (Measurement)
2. ASC/N1483: Basic of Automobile Technology
3. ASC/N1484: Tools & Equipment
4. ASC/N1486: Workshop Safety
5. ASC/N1487: Automobile Electrical System
6. ASC/N1488: Automobile Electronics
7. ASC/N1489: Advance Automobile Technology
8. ASC/N9835: Applied Mathematics
9. DGT/VSQ/N0104: Employability Skills (120 Hours)
10. ASC/N1492: Workshop Technology Two-Wheeler (OJT)

Options*(Not mandatory):*

Option 1: Semester 1: Two-Wheeler Technology

1. ASC/N1485: Two-Wheeler Technology

Option 2: Semester 1: Engineering Drawing

1. ASC/N6458: Engineering Drawing

Option 3: Semester 2: Electric Two or Three Wheeler Technology

1. ASC/N1490: Electric Two or Three Wheeler Technology

Option 4: Semester 2: Four-Wheeler Technology

1. ASC/N1491: Four-Wheeler Technology

Qualification Pack (QP) Parameters

Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
Country	India
NSQF Level	3.5
Credits	46
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7231.0101
Minimum Educational Qualification & Experience	10th Class
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	15 Years
Last Reviewed On	NA
Next Review Date	15/03/2027
NSQC Approval Date	15/03/2023
Version	1.0
Reference code on NQR	QG-3.5-AU-02195-2024-V1-ASDC
NQR Version	1

Remarks:

Mandatory: It is Mandatory to select at least one optional NOS in every semester to meet the 40 credits requirement in a year for diploma progression (As per NCVET Diploma guidelines)

ASC/N6314: Metrology (Measurement)

Description

This NOS unit is about to Employ Metrology Techniques & Tools to inspect the Quality.

Scope

The scope covers the following :

- Identifying the metrology requirements
- Selecting appropriate metrology tools and techniques
- Performing measurements

Elements and Performance Criteria

Identifying the metrology requirements

To be competent, the user/individual on the job must be able to:

PC1. identify various measuring tools and instruments, such as calipers, micrometers, and coordinate measuring machines (CMMs), to verify the dimensions and tolerances of the manufactured components.

PC2. Ensuring that the manufactured parts meet the specified dimensional requirements is crucial.

PC3. conduct regular checks of the machines with support of the maintenance & Production team to identify Quality Issues.

PC4. ensure that all the tools/equipment/fasteners/spare parts are arranged as per specifications/utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/work instructions.

Selecting appropriate metrology tools and techniques

To be competent, the user/individual on the job must be able to:

PC5. Research and select the appropriate metrology tools, such as calipers, micrometers, or coordinate measuring machines (CMMs), based on the type of measurements needed and the precision required.

PC6. Determine the most suitable metrology techniques for the specific task, such as dimensional inspection, surface finish analysis, or geometric tolerance verification.

PC7. Ensure that the selected tools are regularly calibrated and well-maintained to guarantee accurate measurements.

PC8. Maintain accurate records of measurements, including any deviations from specified tolerances, and report them to the relevant stakeholders.

Performing measurements

To be competent, the user/individual on the job must be able to:

PC9. perform quality checks to ensure that the final assembly meets the required metrology standards

PC10. prepare and analyze material and energy audit reports to decipher excessive consumption of material and water.

PC11. Regularly calibrate and maintain metrology tools to ensure their accuracy and reliability.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. organisation procedures for health, safety and security, individual role and responsibilities in this context
KU2. the organisation's emergency procedures for different emergency situations and the importance of following the same
KU3. evacuation procedures for workers and visitors
KU4. how and when to report hazards as well as the limits of responsibility for dealing with hazards
KU5. potential hazards, risks and threats based on the nature of work
KU6. various types of fire extinguisher
KU7. various types of safety signs and their meaning
KU8. Familiarity with various measuring instruments: A Machining and Assembly Technician should know how to use different measuring tools such as calipers, micrometers, dial indicators, and coordinate measuring machines (CMMs).
KU9. Understanding of geometric dimensions and tolerances (GD&T): GD&T is a language used to define the form, orientation, and location of features on a part. A technician must be able to interpret and apply GD&T symbols to ensure the correct assembly and function of the product.
KU10. Knowledge of quality control and assurance principles: Understanding the importance of maintaining quality standards and implementing quality control measures is essential for a Machining and Assembly Technician.
KU11. Ability to perform inspection and measurement tasks: A technician should be able to perform measurements, compare them with the required specifications, and make necessary adjustments to the manufacturing process to maintain quality.
KU12. Familiarity with statistical process control (SPC): SPC helps to monitor and control the manufacturing process by collecting and analyzing data. A Machining and Assembly Technician should be able to use SPC techniques to identify trends and make improvements in the process.
KU13. Basic understanding of computer-aided design (CAD) and computer-aided manufacturing (CAM): Familiarity with these technologies can help a technician better understand the design intent and make informed decisions during the inspection and assembly processes.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read instructions/guidelines/procedures
GS2. modify work practices to improve them
GS3. work with supervisors/team members to carry out work related tasks
GS4. complete tasks efficiently and accurately within stipulated time

GS5. inform/report to concerned person in case of any problem

GS6. make timely decisions for efficient utilization of resources

GS7. write reports such as accident report, in at least English/regional language

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Identifying the metrology requirements	**10**	**20**	**20**	**-**
PC1. identify various measuring tools and instruments, such as calipers, micrometers, and coordinate measuring machines (CMMs), to verify the dimensions and tolerances of the manufactured components.	3	5	5	-
PC2. Ensuring that the manufactured parts meet the specified dimensional requirements is crucial.	3	5	5	-
PC3. conduct regular checks of the machines with support of the maintenance & Production team to identify Quality Issues.	2	5	5	-
PC4. ensure that all the tools/equipment/fasteners/spare parts are arranged as per specifications/utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/work instructions.	2	5	5	-
Selecting appropriate metrology tools and techniques	**5**	**10**	**10**	**-**
PC5. Research and select the appropriate metrology tools, such as calipers, micrometers, or coordinate measuring machines (CMMs), based on the type of measurements needed and the precision required.	1	2	2	-
PC6. Determine the most suitable metrology techniques for the specific task, such as dimensional inspection, surface finish analysis, or geometric tolerance verification.	2	3	3	-
PC7. Ensure that the selected tools are regularly calibrated and well-maintained to guarantee accurate measurements.	1	2	2	-
PC8. Maintain accurate records of measurements, including any deviations from specified tolerances, and report them to the relevant stakeholders.	1	3	3	-
Performing measurements	**5**	**10**	**10**	**-**
PC9. perform quality checks to ensure that the final assembly meets the required metrology standards	2	4	4	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. prepare and analyze material and energy audit reports to decipher excessive consumption of material and water.	2	3	3	-
PC11. Regularly calibrate and maintain metrology tools to ensure their accuracy and reliability.	1	3	3	-
NOS Total	**20**	**40**	**40**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N6314
NOS Name	Metrology (Measurement)
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Automotive Quality Assurance
NSQF Level	3.5
Credits	4
Version	1.0
Last Reviewed Date	15/03/2023
Next Review Date	15/03/2026
NSQC Clearance Date	15/03/2023

ASC/N1483: Basic of Automobile Technology

Description

This NOS unit is about the practice of interpreting basic automobile technologies in the field of service and maintenance to .

Scope

The scope covers the following :

- Diagnose problems effectively.
- Pinpoint the root cause of issues and recommend appropriate repairs or maintenance procedures.
- Perform Functional Checks
- Conclude the repair solution for the fault

Elements and Performance Criteria

Diagnose problems effectively

To be competent, the user/individual on the job must be able to:

PC1. Analyze data from diagnostic tools and identify the root cause of problems efficiently

PC2. List out the customer's description of any problems or concerns they have noticed when driving the vehicle

PC3. Conduct a visual inspection of the exterior and interior of the vehicle to identify any visible signs of damage, wear, or deterioration

Pinpoint the root cause of issues and recommend appropriate repairs or maintenance procedures.

To be competent, the user/individual on the job must be able to:

PC4. Determine the precise location of faults in vehicle systems

PC5. Check for dents, scratches, rust, leaks, fluid stains, tire wear, and other observable issues.

PC6. Conduct test drive to check vehicle performance and identify/validate the faults

PC7. Assess mechanical components/aggregates such as brake pad/shoe, wheel cylinder, mountings, bushes, control arms etc. of the vehicle for any external impact/bend/leak/incorrect level/wear & tear

PC8. Collect workshop tools/measuring device/equipment required for fault diagnosis in vehicle systems and check their condition/calibration

PC9. Break down the vehicle systems (engine, transmission, suspension, brakes, etc.) and analyze each one systematically.

Perform Functional Checks

To be competent, the user/individual on the job must be able to:

PC10. Perform functional checks of critical systems and components, including lights, signals, horn, wipers, windows, doors, locks, and mirrors, to ensure proper operation.

PC11. Use checklists and OEM Standard Operating Procedures (SOPs) to understand if the fault is because of improper charging, loose/poor contacts of pins in wiring harness connectors and their connection, improper driving style etc.

PC12. Select and apply the appropriate device/equipment to make inspection and diagnose deficiencies/faults in various systems such as Body management systems, Braking and stability control systems, Drive line systems, Battery management system, Telematics system, etc.

PC13. Utilize diagnostic tools and equipment to conduct comprehensive diagnostic tests on the vehicle's onboard systems, including engine, transmission, brakes, suspension, steering, and electrical systems

PC14. Follow SOP set out for troubleshooting and perform tests using various mechanical, electrical/electronic measuring devices/tester/diagnostic tools/software to identify/isolate a fault

Conclude the repair solution for the fault

To be competent, the user/individual on the job must be able to:

PC15. maintain the documentation related to inspections and troubleshooting performed on the vehicle

PC16. Compare results of diagnostic inspections/tests with vehicle specifications and regulatory requirements

PC17. Validate the options for repair or replacement

PC18. Prepare final proposal regarding repair or replacement requirements with justification

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The automotive industry workshop structure and role and responsibilities of different people in the workshop

KU2. SOP for receiving vehicles, opening job card, allocation of work, invoicing, vehicle delivery, handling complaints etc

KU3. Different components/aggregates as well as auto component manufacturer's specifications for the same

KU4. basic technology used in and functioning of various systems and components of the vehicle such as batteries, body management system, telematics, brake system, air-conditioning systems, active & passive safety system, media and other systems (including electrical machines and devices used in electric vehicles such as: generator, DC/AC and DC/DC converters, AC motor, DC motor, charging systems etc.)

KU5. Interconnection of systems with each other and effect of one system on other system

KU6. fundamental terms, laws and principles of electricity used in EV such as: principles of storing electrical voltage, ohms law, voltage, current (AC/DC/HV), resistance, power, capacitance, electrostatics, magnetic, inductance, discrete electronic components, and radio frequency (automotive digital computers, automotive communication protocols such as CAN, LIN, etc.)

KU7. use of relevant measuring device/equipment and interpretation of all relevant mathematical calculations

KU8. various electrical and electronic signals such as electrical inputs, outputs, voltage, pulse-width modulation, digital signal (including infra red and fiber optics) etc.

KU9. symbols, units and terms used in wiring diagrams associated with electrical/electric systems/components of the vehicle

KU10. how to use computer, on-line application and OEM technical information/assistance portals

KU11. various sources of information available for assessing service and repair requirements of the vehicle including diagnostic displays, visual inspections, test drives, vehicle/equipment manufacturer specifications, and tolerance limits of components

KU12. typical symptoms of common faults and failures in vehicle mechanical, electrical and electronic systems

KU13. Safety, health and environmental policies and regulations for the work place as well as for automotive trade in general

KU14. Standard Operating Procedures (SOPs) of the organization/ dealership for inspection and diagnosis of faults in a vehicle as prescribed by the OEM/components manufacturer

KU15. How to work on the HV systems which do not require isolation, troubleshooting and replacing parts on the active HV system

KU16. SOP recommended by OEM for using tools/equipment for diagnosis or troubleshooting such as special service tools, measuring instrument, volt meters, ammeters, ohmmeters, battery tester, dedicated and computer based diagnostic equipment, oscilloscopes etc.

KU17. Documentation requirements for each procedure carried out as part of roles and responsibilities as specified by OEM/ auto component manufacturer

KU18. Organizational/professional code of ethics and standards of practice.

KU19. Electrical hazards, protective measures and first aid: in case of electric shock, electrical arc in public grid or in an electrical car, impact of electric current/arc, secondary accidents.

KU20. Five safety rules for electrical work on HV systems before starting the work i.e. isolate, safeguard reconnection, verify the non-live state, earth or short-circuit and shroud or safeguard adjacent live parts.

KU21. Safety requirements recommended by the OEM for equipment /vehicle components during diagnosis, troubleshooting and root cause analysis on various aggregates.

KU22. Legal regulations that need to be considered for handling electric vehicles in the workshop

KU23. Occupational Safety and Health (OSH) measures required for working on electric vehicles

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret workplace related documentation

GS2. communicate using terms, names, grades and other nomenclature pertaining to the automotive trade

GS3. analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently

GS4. identify potential workplace problem and take suitable action

GS5. read various sources of information available for assessing service and repair requirements

GS6. write in English/regional language

GS7. read policies and regulations pertinent to the job, including OEM guidelines, Health and Safety instructions etc. while working on the Electric Vehicle and its aggregates

GS8. communicate effectively at the workplace

GS9. plan work according to the required schedule and location

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Diagnose problems effectively	**5**	**10**	**10**	**-**
PC1. Analyze data from diagnostic tools and identify the root cause of problems efficiently	-	-	-	-
PC2. List out the customer's description of any problems or concerns they have noticed when driving the vehicle	-	-	-	-
PC3. Conduct a visual inspection of the exterior and interior of the vehicle to identify any visible signs of damage, wear, or deterioration	-	-	-	-
Pinpoint the root cause of issues and recommend appropriate repairs or maintenance procedures.	**5**	**10**	**10**	**-**
PC4. Determine the precise location of faults in vehicle systems	-	-	-	-
PC5. Check for dents, scratches, rust, leaks, fluid stains, tire wear, and other observable issues.	-	-	-	-
PC6. Conduct test drive to check vehicle performance and identify/validate the faults	-	-	-	-
PC7. Assess mechanical components/aggregates such as brake pad/shoe, wheel cylinder, mountings, bushes, control arms etc. of the vehicle for any external impact/bend/leak/incorrect level/wear & tear	-	-	-	-
PC8. Collect workshop tools/measuring device/equipment required for fault diagnosis in vehicle systems and check their condition/calibration	-	-	-	-
PC9. Break down the vehicle systems (engine, transmission, suspension, brakes, etc.) and analyze each one systematically.	-	-	-	-
Perform Functional Checks	**5**	**10**	**10**	**-**
PC10. Perform functional checks of critical systems and components, including lights, signals, horn, wipers, windows, doors, locks, and mirrors, to ensure proper operation.	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Use checklists and OEM Standard Operating Procedures (SOPs) to understand if the fault is because of improper charging, loose/poor contacts of pins in wiring harness connectors and their connection, improper driving style etc.	-	-	-	-
PC12. Select and apply the appropriate device/equipment to make inspection and diagnose deficiencies/faults in various systems such as Body management systems, Braking and stability control systems, Drive line systems, Battery management system, Telematics system, etc.	-	-	-	-
PC13. Utilize diagnostic tools and equipment to conduct comprehensive diagnostic tests on the vehicle's onboard systems, including engine, transmission, brakes, suspension, steering, and electrical systems	-	-	-	-
PC14. Follow SOP set out for troubleshooting and perform tests using various mechanical, electrical/electronic measuring devices/tester/diagnostic tools/software to identify/isolate a fault	-	-	-	-
Conclude the repair solution for the fault	**5**	**10**	**10**	**-**
PC15. maintain the documentation related to inspections and troubleshooting performed on the vehicle	-	-	-	-
PC16. Compare results of diagnostic inspections/tests with vehicle specifications and regulatory requirements	-	-	-	-
PC17. Validate the options for repair or replacement	-	-	-	-
PC18. Prepare final proposal regarding repair or replacement requirements with justification	-	-	-	-
NOS Total	**20**	**40**	**40**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1483
NOS Name	Basic of Automobile Technology
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	3.5
Credits	1.5
Version	1.0
Last Reviewed Date	15/03/2023
Next Review Date	15/03/2026
NSQC Clearance Date	15/03/2023

ASC/N1484: Tools & Equipment

Description

This unit describes the knowledge and skills required in an individual to use the tools and equipment inside a workshop and carry out service, repair and overhauling of mechanical system of an automobile.

Scope

The scope covers the following :

- Using diagnostic tools to perform routine service and repairs.
- Perform service and repair with hand tools.
- Initiating service/repair with power tools.
- Operating with lifting Equipment.

Elements and Performance Criteria

Using diagnostic tools to perform routine service and repairs.

To be competent, the user/individual on the job must be able to:

PC1. Access real-time data streams from various sensors and modules within the vehicle.

PC2. Perform routine health checks on vehicles using diagnostic tools like OBD-II scanners.

PC3. Scan the vehicle's onboard computer systems to detect any fault codes or irregularities in various components such as the engine, transmission, brakes, and emissions systems.

PC4. Collect workshop tools/measuring devices/equipment required for the job and check their condition/calibration.

PC5. Report the malfunctions if any, in the tools/equipment to the concerned personnel,

Perform service and repair with hand tools

To be competent, the user/individual on the job must be able to:

PC6. Hand tools such as screwdrivers, wrenches, and sockets to be used to disassemble and reassemble various components of the vehicle, including interior panels, engine covers, and exterior trim

PC7. Slicing and trimming with the help of wire cutters, crimping tools, and strippers for electrical work, including repairing wiring harnesses, splicing wires, and installing electrical components.

PC8. Tightening bolts using pliers, wrenches, and sockets to work on fluid systems, including the engine's cooling system, fuel system, and brake system.

PC9. Tapping with hammers, drift punches, and bearing drivers to install bearings, seals, and bushings.

PC10. Carefully tapping these components into place to ensure proper seating without causing damage.

Initiating service/repair with power tools

To be competent, the user/individual on the job must be able to:

PC11. Impact wrenches are used to quickly and easily tighten or loosen nuts and bolts, especially for tasks such as wheel removal and suspension work.

PC12. High torque output of impact wrenches allows technicians to complete tasks faster than with manual wrenches.

PC13. Air compressors power a variety of pneumatic tools used by technicians, including impact wrenches, air ratchets, air hammers, and pneumatic drills

PC14. Electric drills and drivers are used for drilling holes, driving screws, and fastening bolts and nuts.

PC15. Sanders and polishers are used for smoothing surfaces, removing paint or clear coat, and restoring shine to painted surfaces

Operating lifting Equipment.

To be competent, the user/individual on the job must be able to:

PC16. Hydraulic lifts or vehicle hoists are used to elevate vehicles to a comfortable working height for thorough inspection and diagnosis.

PC17. Portable devices like hydraulic floor jacks are used to lift vehicles off the ground for tire changes, brake servicing, and other minor repairs

PC18. Four-post lifts have four vertical posts with a platform on top are often used for wheel alignment, undercarriage inspections, and long-term storage.

PC19. Alignment racks are used for performing wheel alignments which allow technicians to adjust camber, caster, and toe angles while the vehicle is lifted off the ground

PC20. Two-post lifts whic have two upright columns with adjustable arms that lift the vehicle by its frame or lifting points.They are commonly used for routine maintenance, tire changes, and suspension work.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Technicians should have a comprehensive understanding of various tools and equipment used in automotive repair and maintenance. knowing the names, functions, and proper usage of tools such as wrenches, sockets, screwdrivers, and diagnostic equipment

KU2. Knowing the names, functions, and proper usage of tools such as wrenches, sockets, screwdrivers, and diagnostic equipment is important.

KU3. Understanding which tool is appropriate for a given task is essential to ensure the job is done correctly and efficiently.

KU4. Must know the differences between various types and sizes of tools and select the most suitable one for the job at hand

KU5. Knowledge of safety precautions when using tools and equipment is paramount to prevent accidents and injuries

KU6. Understanding proper lifting techniques, wearing appropriate personal protective equipment (PPE), and following safety guidelines provided by equipment manufacturers.

KU7. various sources of information available for assessing service and repair requirements of the public grid or in an electrical car, impact of electric current/arc, secondary accidents, etc

KU8. Should know how to properly maintain and care for their tools and equipment to ensure their longevity and optimal performance

KU9. Practicing cleaning, lubricating, and storing tools correctly, as well as performing regular inspections for wear and damage

KU10. For diagnostic equipment such as scan tools and multimeters, technicians should have a thorough understanding of how to operate them.

KU11. Knowledge of interpreting diagnostic trouble codes (DTCs), navigating through system menus, and analyzing data to diagnose vehicle issues accurately

KU12. Should be knowledgeable about the calibration process and ensure that equipment is calibrated according to manufacturer specifications.

KU13. Understanding how to troubleshoot equipment malfunctions or tool failures is essential for minimizing downtime and maintaining productivity in the shop

KU14. Should be able to identify common issues and perform basic troubleshooting steps to resolve them

KU15. Must stay updated with the latest tools and equipment

KU16. Commitment to continuous learning through training programs, workshops, and industry publications to enhance knowledge and skills

KU17. Should maintain proper documentation of tool usage, equipment maintenance, and diagnostic procedures. This helps track equipment performance, identify trends, and ensure compliance with regulatory requirements

KU18. Should be willing to share knowledge and expertise with colleagues regarding tool usage and best practices. This fosters a culture of learning and continuous improvement within the automotive service facility

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret workplace related documentation

GS2. interpret the needs of customers by understanding the key issues

GS3. communicate using terms, names, grades and other nomenclature pertaining to the automotive trade

GS4. analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently

GS5. identify potential workplace problem and take suitable action

GS6. write in English/regional language

GS7. read various sources of information available for assessing service and repair requirements

GS8. read policies and regulations pertinent to the job, including OEM guidelines, Health and Safety instructions etc. while working on the Electric Vehicle and its aggregates

GS9. communicate effectively at the workplace

GS10. plan work according to the required schedule and location

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Using diagnostic tools to perform routine service and repairs.	**5**	**10**	**10**	**-**
PC1. Access real-time data streams from various sensors and modules within the vehicle.	-	-	-	-
PC2. Perform routine health checks on vehicles using diagnostic tools like OBD-II scanners.	-	-	-	-
PC3. Scan the vehicle's onboard computer systems to detect any fault codes or irregularities in various components such as the engine, transmission, brakes, and emissions systems.	-	-	-	-
PC4. Collect workshop tools/measuring devices/equipment required for the job and check their condition/calibration.	-	-	-	-
PC5. Report the malfunctions if any, in the tools/equipment to the concerned personnel,	-	-	-	-
Perform service and repair with hand tools	**5**	**10**	**10**	**-**
PC6. Hand tools such as screwdrivers, wrenches, and sockets to be used to disassemble and reassemble various components of the vehicle, including interior panels, engine covers, and exterior trim	-	-	-	-
PC7. Slicing and trimming with the help of wire cutters, crimping tools, and strippers for electrical work, including repairing wiring harnesses, splicing wires, and installing electrical components.	-	-	-	-
PC8. Tightening bolts using pliers, wrenches, and sockets to work on fluid systems, including the engine's cooling system, fuel system, and brake system.	-	-	-	-
PC9. Tapping with hammers, drift punches, and bearing drivers to install bearings, seals, and bushings.	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Carefully tapping these components into place to ensure proper seating without causing damage.	-	-	-	-
Initiating service/repair with power tools	**5**	**10**	**10**	**-**
PC11. Impact wrenches are used to quickly and easily tighten or loosen nuts and bolts, especially for tasks such as wheel removal and suspension work.	-	-	-	-
PC12. High torque output of impact wrenches allows technicians to complete tasks faster than with manual wrenches.	-	-	-	-
PC13. Air compressors power a variety of pneumatic tools used by technicians, including impact wrenches, air ratchets, air hammers, and pneumatic drills	-	-	-	-
PC14. Electric drills and drivers are used for drilling holes, driving screws, and fastening bolts and nuts.	-	-	-	-
PC15. Sanders and polishers are used for smoothing surfaces, removing paint or clear coat, and restoring shine to painted surfaces	-	-	-	-
Operating lifting Equipment.	**5**	**10**	**10**	**-**
PC16. Hydraulic lifts or vehicle hoists are used to elevate vehicles to a comfortable working height for thorough inspection and diagnosis.	-	-	-	-
PC17. Portable devices like hydraulic floor jacks are used to lift vehicles off the ground for tire changes, brake servicing, and other minor repairs	-	-	-	-
PC18. Four-post lifts have four vertical posts with a platform on top are often used for wheel alignment, undercarriage inspections, and long-term storage.	-	-	-	-
PC19. Alignment racks are used for performing wheel alignments which allow technicians to adjust camber, caster, and toe angles while the vehicle is lifted off the ground	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC20. Two-post lifts whic have two upright columns with adjustable arms that lift the vehicle by its frame or lifting points.They are commonly used for routine maintenance, tire changes, and suspension work.	-	-	-	-
NOS Total	**20**	**40**	**40**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1484
NOS Name	Tools & Equipment
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	3.5
Credits	1.5
Version	1.0
Last Reviewed Date	15/03/2023
Next Review Date	15/03/2026
NSQC Clearance Date	15/03/2023

ASC/N1486: Workshop Safety

Description

This unit describes the importance of Adherence to workshop safety guidelines.

Scope

The scope covers the following :

- Adherence to Safety Protocols and Procedures.
- Performing regular Equipment Maintenance and Inspection.
- Communication and Collaboration.

Elements and Performance Criteria

Adherence to Safety Protocols and Procedures

To be competent, the user/individual on the job must be able to:

PC1. Following safety protocols and procedures to prevent accidents and injuries by identifying and mitigating potential hazards in the workshop.

PC2. Wearing appropriate personal protective equipment (PPE), employing safe work practices, and following established procedures for handling tools, equipment, and hazardous materials, can reduce the risk of workplace accidents

PC3. Adhering to safety protocols ensures compliance with regulations and standards set by occupational safety agencies and other regulatory bodies.

PC4. Compliance with these regulations helps protect technicians from legal liabilities and ensures the workshop operates in accordance with industry best practices

PC5. By following established safety protocols, technicians reduce the likelihood of encountering work-related risks such as chemical exposure, electrical shock, ergonomic injuries, and accidents involving heavy machinery or vehicles.

Performing Regular Equipment Maintenance and Inspection

To be competent, the user/individual on the job must be able to:

PC6. Conduct regular maintenance and inspection of workshop equipment, tools, and machinery to ensure they are in proper working condition.

PC7. Regular maintenance helps identify and address any issues or wear and tear in equipment before they lead to failures during operation.

PC8. Timely Inspections prevent unexpected breakdowns that could potentially cause accidents or injuries

Communication and Collaboration

To be competent, the user/individual on the job must be able to:

PC9. Well-maintained equipment operates more efficiently and effectively. This reduces the likelihood of malfunctions or errors that could pose safety risks to technicians or damage to vehicles

PC10. Effective communication ensures that all technicians are aware of safety procedures, protocols, and potential hazards in the workshop.

PC11. Conducting regular safety meetings where updates, reminders, and incident reports are discussed should be brought into practice to keep everyone informed

PC12. Through open communication and collaboration, technicians can actively participate in the continuous improvement of safety practices in the workshop

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. safety, health and environmental policies and regulations of the workplace as well as for automotive trade in general (e.g. safe practices while working in pits/under vehicles)

KU2. Understanding of safety systems found in LCVs, including airbag systems, seatbelt pretensioners, and collision avoidance systems. Knowledge of tire technology.

KU3. Safety precautions for equipment and components prescribed by the OEM.

KU4. Various methods to dispose-off replaced failed components/parts in accordance with safety, health and environmental policies and regulation.

KU5. Occupational Safety and Health (OSH) measures required for working on electric vehicle

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret workplace related documentation

GS2. interpret the needs of customers by understanding the key issues

GS3. communicate using terms, names, grades and other nomenclature pertaining to the automotive trade

GS4. analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently

GS5. identify potential workplace problem and take suitable action

GS6. write in English/regional language

GS7. read various sources of information available for assessing service and repair requirements

GS8. Read policies and regulations pertinent to the job, including OEM guidelines, Health and Safety instructions etc. while working on the Electric Vehicle and its aggregates

GS9. Communicate effectively at the workplace

GS10. Plan work according to the required schedule and location

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Adherence to Safety Protocols and Procedures	**5**	**10**	**10**	**-**
PC1. Following safety protocols and procedures to prevent accidents and injuries by identifying and mitigating potential hazards in the workshop.	-	-	-	-
PC2. Wearing appropriate personal protective equipment (PPE), employing safe work practices, and following established procedures for handling tools, equipment, and hazardous materials, can reduce the risk of workplace accidents	-	-	-	-
PC3. Adhering to safety protocols ensures compliance with regulations and standards set by occupational safety agencies and other regulatory bodies.	-	-	-	-
PC4. Compliance with these regulations helps protect technicians from legal liabilities and ensures the workshop operates in accordance with industry best practices	-	-	-	-
PC5. By following established safety protocols, technicians reduce the likelihood of encountering work-related risks such as chemical exposure, electrical shock, ergonomic injuries, and accidents involving heavy machinery or vehicles.	-	-	-	-
Performing Regular Equipment Maintenance and Inspection	**5**	**10**	**10**	**-**
PC6. Conduct regular maintenance and inspection of workshop equipment, tools, and machinery to ensure they are in proper working condition.	-	-	-	-
PC7. Regular maintenance helps identify and address any issues or wear and tear in equipment before they lead to failures during operation.	-	-	-	-
PC8. Timely Inspections prevent unexpected breakdowns that could potentially cause accidents or injuries	-	-	-	-
Communication and Collaboration	**5**	**10**	**10**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. Well-maintained equipment operates more efficiently and effectively. This reduces the likelihood of malfunctions or errors that could pose safety risks to technicians or damage to vehicles	-	-	-	-
PC10. Effective communication ensures that all technicians are aware of safety procedures, protocols, and potential hazards in the workshop.	-	-	-	-
PC11. Conducting regular safety meetings where updates, reminders, and incident reports are discussed should be brought into practice to keep everyone informed	-	-	-	-
PC12. Through open communication and collaboration, technicians can actively participate in the continuous improvement of safety practices in the workshop	-	-	-	-
NOS Total	**15**	**30**	**30**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1486
NOS Name	Workshop Safety
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	3.5
Credits	5
Version	1.0
Last Reviewed Date	15/03/2023
Next Review Date	15/03/2026
NSQC Clearance Date	15/03/2023

ASC/N1487: Automobile Electrical System

Description

This unit describes the knowledge and understanding required by an individual to diagnose electrical systems in an automobile effectively.

Scope

The scope covers the following :

- Conduct battery testing.
- Perform Starting System Diagnostics.
- Conduct Lighting System Diagnostic.
- Troubleshoot Ignition Systems.

Elements and Performance Criteria

Conduct battery testing

To be competent, the user/individual on the job must be able to:

PC1. Begin by visually inspecting the battery for any signs of physical damage, corrosion on terminals, or leaks.

PC2. Look for cracks, bulges, or discoloration on the battery casing, which may indicate internal damage

PC3. Use a multimeter to measure the battery's voltage. With the engine off and all accessories turned off, connect the multimeter's positive lead to the positive terminal of the battery and the negative lead to the negative terminal

PC4. Perform a load test to assess the battery's ability to deliver power under load. Use a battery load tester to apply a load equivalent to half the battery's CCA (Cold Cranking Amps) rating for 15 seconds while monitoring the voltage

PC5. Measure the specific gravity of the electrolyte using a hydrometer for flooded lead-acid batteries

Perform Starting System Diagnostics

To be competent, the user/individual on the job must be able to:

PC6. Check for audible clicking noises when attempting to start the vehicle. If present, it could indicate a faulty starter solenoid.

PC7. Inspect the starter motor for physical damage, such as worn brushes, damaged windings, or corrosion.

PC8. Test the starter motor's electrical connections for corrosion or looseness, ensuring a secure and clean connection.

PC9. Verify that the ignition switch is functioning correctly by checking for power at the starter solenoid when the key is turned to the start position

PC10. Test the starter relay or solenoid for proper operation by bypassing it with a jumper wire and observing if the starter engages

PC11. Check for voltage drop between the battery and starter motor during cranking. Excessive voltage drop indicates resistance in the circuit, potentially caused by corroded or loose connections.

PC12. Inspect the wiring harness for any signs of damage, fraying, or breaks, especially in the vicinity of the starter motor and solenoid.

Conduct Lighting System Diagnostic

To be competent, the user/individual on the job must be able to:

PC13. Visually inspect all exterior and interior lights, including headlights, taillights, turn signals, brake lights, and interior illumination.

PC14. Look for burned-out bulbs, cracked lenses, loose connections, or signs of corrosion on the light assemblies and wiring

PC15. Test each lighting function individually by activating the corresponding switches or controls.

PC16. Verify that each light illuminates properly and performs its intended function, such as low beams, high beams, turn signals, hazard lights, and brake lights

PC17. Use a multimeter to check voltage and continuity at various points in the lighting circuit, including the battery, fuses, relays, switches, and light sockets.

PC18. Test for voltage drop across connections and wiring harnesses to identify high-resistance areas that may cause dim or non-functional lights

Troubleshoot Ignition Systems

To be competent, the user/individual on the job must be able to:

PC19. The first step in diagnosing an ignition system is to determine if the spark plugs are firing properly

PC20. This can be done by removing a spark plug wire and grounding it to the engine block while cranking the engine. If there's a visible spark, it indicates that the ignition system is producing sufficient voltage

PC21. Examine the condition of the spark plugs for signs of fouling, wear, or damage. Fouled or worn- out spark plugs can lead to misfires and poor engine performance. Different types of deposits on the spark plugs can provide clues about engine condition and potential problems.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The automotive Industry, workshop structure and role and responsibilities of different people in the workshop

KU2. Understanding basic electrical principles such as voltage, current, resistance, and Ohm's Law

KU3. Knowledge of electrical circuits, including series circuits, parallel circuits, and series-parallel circuits.

KU4. Understanding battery types (e.g., lead-acid, AGM, lithium-ion) and their characteristics

KU5. Knowledge of battery construction, including cells, terminals, and electrolyte

KU6. Understanding battery charging systems, including alternators, voltage regulators, and charging circuits

KU7. Knowledge of battery testing methods, such as load testing and specific gravity testing

KU8. Understanding the components of the charging system, including alternators, voltage regulators, and diodes

KU9. Knowledge of alternator operation, including AC generation, rectification, and voltage regulation

KU10. Ability to diagnose charging system issues, such as low output voltage, overcharging, or charging system warning lights

KU11. Understanding the components of the starting system, including starters, solenoids, ignition switches, and relays

KU12. Ability to diagnose starting system issues, such as no-crank conditions, starter motor failure, or ignition switch faults

KU13. Understanding the components of the ignition system, including ignition coils, spark plugs, ignition control modules, and sensors

KU14. Knowledge of ignition system types, including distributor-based systems and distributorless ignition systems (DIS).

KU15. Ability to diagnose ignition system issues, such as misfires, weak spark, or ignition timing problems

KU16. Understanding lighting system components, including bulbs, fuses, relays, switches, and wiring

KU17. Knowledge of different types of lighting systems, such as incandescent, halogen, LED, and HID

KU18. Ability to diagnose lighting system issues, such as non-functional lights, flickering lights, or electrical shorts

KU19. Ability to read and interpret wiring diagrams and schematics to understand vehicle electrical circuits

KU20. Familiarity with diagnostic tools such as multimeters, test lights, circuit testers, and scan tools. Knowledge of how to use diagnostic tools to perform electrical tests, retrieve diagnostic trouble codes (DTCs), and interpret data

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret workplace related documentation

GS2. interpret the needs of customers by understanding the key issues

GS3. communicate using terms, names, grades and other nomenclature pertaining to the automotive trade

GS4. analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently

GS5. identify potential workplace problem and take suitable action

GS6. write in English/regional language

GS7. read various sources of information available for assessing service and repair requirements

GS8. Read policies and regulations pertinent to the job, including OEM guidelines, Health and Safety instructions etc. while working on the Electric Vehicle and its aggregates

GS9. Communicate effectively at the workplace

GS10. Plan work according to the required schedule and location

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Conduct battery testing	**5**	**10**	**10**	**-**
PC1. Begin by visually inspecting the battery for any signs of physical damage, corrosion on terminals, or leaks.	-	-	-	-
PC2. Look for cracks, bulges, or discoloration on the battery casing, which may indicate internal damage	-	-	-	-
PC3. Use a multimeter to measure the battery's voltage. With the engine off and all accessories turned off, connect the multimeter's positive lead to the positive terminal of the battery and the negative lead to the negative terminal	-	-	-	-
PC4. Perform a load test to assess the battery's ability to deliver power under load. Use a battery load tester to apply a load equivalent to half the battery's CCA (Cold Cranking Amps) rating for 15 seconds while monitoring the voltage	-	-	-	-
PC5. Measure the specific gravity of the electrolyte using a hydrometer for flooded lead-acid batteries	-	-	-	-
Perform Starting System Diagnostics	**5**	**10**	**10**	**-**
PC6. Check for audible clicking noises when attempting to start the vehicle. If present, it could indicate a faulty starter solenoid.	-	-	-	-
PC7. Inspect the starter motor for physical damage, such as worn brushes, damaged windings, or corrosion.	-	-	-	-
PC8. Test the starter motor's electrical connections for corrosion or looseness, ensuring a secure and clean connection.	-	-	-	-
PC9. Verify that the ignition switch is functioning correctly by checking for power at the starter solenoid when the key is turned to the start position	-	-	-	-
PC10. Test the starter relay or solenoid for proper operation by bypassing it with a jumper wire and observing if the starter engages	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Check for voltage drop between the battery and starter motor during cranking. Excessive voltage drop indicates resistance in the circuit, potentially caused by corroded or loose connections.	-	-	-	-
PC12. Inspect the wiring harness for any signs of damage, fraying, or breaks, especially in the vicinity of the starter motor and solenoid.	-	-	-	-
Conduct Lighting System Diagnostic	**5**	**10**	**10**	**-**
PC13. Visually inspect all exterior and interior lights, including headlights, taillights, turn signals, brake lights, and interior illumination.	-	-	-	-
PC14. Look for burned-out bulbs, cracked lenses, loose connections, or signs of corrosion on the light assemblies and wiring	-	-	-	-
PC15. Test each lighting function individually by activating the corresponding switches or controls.	-	-	-	-
PC16. Verify that each light illuminates properly and performs its intended function, such as low beams, high beams, turn signals, hazard lights, and brake lights	-	-	-	-
PC17. Use a multimeter to check voltage and continuity at various points in the lighting circuit, including the battery, fuses, relays, switches, and light sockets.	-	-	-	-
PC18. Test for voltage drop across connections and wiring harnesses to identify high-resistance areas that may cause dim or non-functional lights	-	-	-	-
Troubleshoot Ignition Systems	**5**	**10**	**10**	**-**
PC19. The first step in diagnosing an ignition system is to determine if the spark plugs are firing properly	-	-	-	-
PC20. This can be done by removing a spark plug wire and grounding it to the engine block while cranking the engine. If there's a visible spark, it indicates that the ignition system is producing sufficient voltage	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. Examine the condition of the spark plugs for signs of fouling, wear, or damage. Fouled or worn-out spark plugs can lead to misfires and poor engine performance. Different types of deposits on the spark plugs can provide clues about engine condition and potential problems.	-	-	-	-
NOS Total	**20**	**40**	**40**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1487
NOS Name	Automobile Electrical System
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	3.5
Credits	1.5
Version	1.0
Last Reviewed Date	15/03/2023
Next Review Date	15/03/2026
NSQC Clearance Date	15/03/2023

ASC/N1488: Automobile Electronics

Description

This unit describes the knowledge and understanding required by an individual about the technologies employed in automobile electronics

Scope

The scope covers the following :

- Diagnosing and troubleshooting engine control modules (ECMs),
- Examining transmission control modules (TCMs) and electronic control units (ECUs) issues.
- Inspecting various onboard (OBD) systems.

Elements and Performance Criteria

Diagnosing and troubleshooting control modules (ECMs).

To be competent, the user/individual on the job must be able to:

PC1. Using diagnostic tools such as scan tools or code readers to retrieve diagnostic trouble codes (DTCs) stored in the ECM's memory

PC2. Interpreting DTCs to identify specific faults or malfunctions in the engine control system

PC3. Accessing live data streams from the ECM to monitor engine parameters such as engine speed, coolant temperature, throttle position, and oxygen sensor readings

PC4. Analyzing live data to identify abnormalities or discrepancies that may indicate underlying issues with engine performance

PC5. Testing individual components within the engine control system, such as sensors (e.g., throttle position sensor, oxygen sensor) and actuators (e.g., fuel injectors, ignition coils).

PC6. Performing tests such as resistance checks, voltage checks, and signal waveform analysis to verify component functionality.

PC7. Inspecting wiring harnesses, connectors, and grounds associated with the ECM for signs of damage, corrosion, or poor connections.

PC8. Repairing or replacing damaged wiring, connectors, or terminals to ensure proper electrical continuity and signal integrity.

PC9. Updating or reprogramming the ECM's software to address software bugs, improve performance, or address emissions compliance issues

PC10. Calibrating the ECM to optimize engine performance parameters based on vehicle specifications or customer preferences.

Examining transmission control modules (TCMs) and electronic control units (ECUs) issues

To be competent, the user/individual on the job must be able to:

PC11. Using diagnostic scan tools to retrieve diagnostic trouble codes (DTCs) stored in the TCM's/ECU’s memory

PC12. Updating the TCM's/ECU’s software to address known issues, improve transmission performance, or enhance shift quality with the help of manufacturer-specific diagnostic equipment and software to perform software updates and recalibrations

PC13. Conduct electrical tests on TCM circuits, connectors, and wiring harnesses to ensure proper electrical connectivity.

PC14. Check for voltage irregularities, open circuits, short circuits, and other electrical faults that may affect TCM operation.

PC15. Follow manufacturer-specific procedures for TCM/ECU replacement in case of a faulty TCM/ECU, which may include programming or initialization steps to match the new TCM to the vehicle.

PC16. Conduct tests on engine sensors and actuators that interface with the ECU, such as throttle position sensors, oxygen sensors, and fuel injectors.

PC17. Perform performance tuning on the ECU to optimize engine performance parameters such as fuel delivery, ignition timing, and turbo boost pressure. Adjusting these parameters helps to achieve desired performance characteristics, such as increased horsepower or improved fuel efficiency.

PC18. Reset the ECU's adaptive learning values and fuel trim adjustments to baseline values

PC19. Conduct final testing to ensure proper functionality after completing maintenance or repair tasks on the ECU/TCM.

PC20. Verify that DTCs are cleared, sensor readings are within specifications, and the engine operates smoothly under various operating conditions.

Inspecting various onboard (OBD) systems

To be competent, the user/individual on the job must be able to:

PC21. Use OBD scanners to retrieve diagnostic trouble codes (DTCs) stored in the vehicle's computer.

PC22. Interpret the codes to identify specific issues affecting engine performance, emissions, and other vehicle systems.

PC23. Visually inspect components of the OBD system, including wiring harnesses, connectors, and sensors, for signs of damage, corrosion, or loose connections.

PC24. Conduct tests on various sensors connected to the OBD system, such as oxygen sensors, mass airflow sensors, and throttle position sensors.

PC25. Verify sensor readings using diagnostic equipment to ensure they fall within acceptable ranges and respond correctly to changes in vehicle operation.

PC26. Document diagnostic findings, repair procedures, and parts used for maintaining accurate service records.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understanding of basic electrical principles such as voltage, current, resistance, and power

KU2. Understanding basic electrical principles such as voltage, current, resistance, and Ohm's Law.

KU3. Familiarity with electrical components such as wires, connectors, fuses, relays, and switches

KU4. Understanding of ECMs and their role as the central control unit for managing engine performance, emissions, and other vehicle systems

KU5. Knowledge of ECM programming, calibration, and diagnostic procedures using specialized tools and software.

KU6. Familiarity with the components integrated into ECMs, including microprocessors, memory chips, and input/output (I/O) ports.
KU7. Understanding of various sensors used in automotive systems, such as throttle position sensors, oxygen sensors, crankshaft position sensors, and temperature sensors.
KU8. Knowledge of diagnostic trouble codes (DTCs), freeze frame data, and readiness monitors used for diagnosing and repairing vehicle issues.
KU9. Understanding of in-vehicle communication networks such as Controller Area Network (CAN bus), Local Interconnect Network (LIN), and FlexRay.
KU10. Knowledge of EFI diagnostics, fuel system testing, and fuel trim adjustments for optimizing engine performance and emissions.
KU11. Understanding of safety systems such as airbag systems, anti-lock braking systems (ABS), traction control systems (TCS), and electronic stability control (ESC).
KU12. Knowledge of safety system diagnostics, including sensor testing, module programming, and recalibration procedures.
KU13. Understanding the components of the ignition system, including ignition coils, spark plugs, ignition control modules, and sensors. Understanding of entertainment and communication systems such as radios, CD players, navigation systems, Bluetooth connectivity, and voice recognition.
KU14. Knowledge of system components, interfaces, and diagnostic procedures for troubleshooting audio, video, and communication issues.
KU15. Understanding of ADAS technologies such as adaptive cruise control, lane departure warning, forward collision warning, and automatic emergency braking.
KU16. Knowledge of ADAS sensors, cameras, radar, and LiDAR systems, as well as calibration procedures and diagnostics.
KU17. Organizational/professional code of ethics and standards of practice
KU18. Documentation requirements for each procedure carried out as part of roles and responsibilities as specified by OEM/auto component manufacturer.
KU19. Various methods to dispose-off replaced failed components/parts in accordance with safety, health and environmental policies and regulation
KU20. safety, health and environmental policies and regulations for the workplace as well as for automotive trade in general

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret workplace related documentation
GS2. interpret the needs of customers by understanding the key issues
GS3. communicate using terms, names, grades and other nomenclature pertaining to the automotive trade
GS4. analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
GS5. identify potential workplace problem and take suitable action
GS6. write in English/regional language
GS7. read various sources of information available for assessing service and repair requirements

GS8. Read policies and regulations pertinent to the job, including OEM guidelines, Health and Safety instructions etc. while working on the Electric Vehicle and its aggregates.

GS9. Communicate effectively at the workplace

GS10. Plan work according to the required schedule and location

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Diagnosing and troubleshooting control modules (ECMs).	**10**	**20**	**20**	**-**
PC1. Using diagnostic tools such as scan tools or code readers to retrieve diagnostic trouble codes (DTCs) stored in the ECM's memory	-	-	-	-
PC2. Interpreting DTCs to identify specific faults or malfunctions in the engine control system	-	-	-	-
PC3. Accessing live data streams from the ECM to monitor engine parameters such as engine speed, coolant temperature, throttle position, and oxygen sensor readings	-	-	-	-
PC4. Analyzing live data to identify abnormalities or discrepancies that may indicate underlying issues with engine performance	-	-	-	-
PC5. Testing individual components within the engine control system, such as sensors (e.g., throttle position sensor, oxygen sensor) and actuators (e.g., fuel injectors, ignition coils).	-	-	-	-
PC6. Performing tests such as resistance checks, voltage checks, and signal waveform analysis to verify component functionality.	-	-	-	-
PC7. Inspecting wiring harnesses, connectors, and grounds associated with the ECM for signs of damage, corrosion, or poor connections.	-	-	-	-
PC8. Repairing or replacing damaged wiring, connectors, or terminals to ensure proper electrical continuity and signal integrity.	-	-	-	-
PC9. Updating or reprogramming the ECM's software to address software bugs, improve performance, or address emissions compliance issues	-	-	-	-
PC10. Calibrating the ECM to optimize engine performance parameters based on vehicle specifications or customer preferences.	-	-	-	-
Examining transmission control modules (TCMs) and electronic control units (ECUs) issues	**5**	**10**	**10**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Using diagnostic scan tools to retrieve diagnostic trouble codes (DTCs) stored in the TCM's/ECU's memory	-	-	-	-
PC12. Updating the TCM's/ECU's software to address known issues, improve transmission performance, or enhance shift quality with the help of manufacturer-specific diagnostic equipment and software to perform software updates and recalibrations	-	-	-	-
PC13. Conduct electrical tests on TCM circuits, connectors, and wiring harnesses to ensure proper electrical connectivity.	-	-	-	-
PC14. Check for voltage irregularities, open circuits, short circuits, and other electrical faults that may affect TCM operation.	-	-	-	-
PC15. Follow manufacturer-specific procedures for TCM/ECU replacement in case of a faulty TCM/ECU, which may include programming or initialization steps to match the new TCM to the vehicle.	-	-	-	-
PC16. Conduct tests on engine sensors and actuators that interface with the ECU, such as throttle position sensors, oxygen sensors, and fuel injectors.	-	-	-	-
PC17. Perform performance tuning on the ECU to optimize engine performance parameters such as fuel delivery, ignition timing, and turbo boost pressure. Adjusting these parameters helps to achieve desired performance characteristics, such as increased horsepower or improved fuel efficiency.	-	-	-	-
PC18. Reset the ECU's adaptive learning values and fuel trim adjustments to baseline values	-	-	-	-
PC19. Conduct final testing to ensure proper functionality after completing maintenance or repair tasks on the ECU/TCM.	-	-	-	-
PC20. Verify that DTCs are cleared, sensor readings are within specifications, and the engine operates smoothly under various operating conditions.	-	-	-	-
Inspecting various onboard (OBD) systems	**5**	**10**	**10**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. Use OBD scanners to retrieve diagnostic trouble codes (DTCs) stored in the vehicle's computer.	-	-	-	-
PC22. Interpret the codes to identify specific issues affecting engine performance, emissions, and other vehicle systems.	-	-	-	-
PC23. Visually inspect components of the OBD system, including wiring harnesses, connectors, and sensors, for signs of damage, corrosion, or loose connections.	-	-	-	-
PC24. Conduct tests on various sensors connected to the OBD system, such as oxygen sensors, mass airflow sensors, and throttle position sensors.	-	-	-	-
PC25. Verify sensor readings using diagnostic equipment to ensure they fall within acceptable ranges and respond correctly to changes in vehicle operation.	-	-	-	-
PC26. Document diagnostic findings, repair procedures, and parts used for maintaining accurate service records.	-	-	-	-
NOS Total	**20**	**40**	**40**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1488
NOS Name	Automobile Electronics
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	3.5
Credits	1.5
Version	1.0
Last Reviewed Date	15/03/2023
Next Review Date	15/03/2026
NSQC Clearance Date	15/03/2023

ASC/N1489: Advance Automobile Technology

Description

This unit describes the knowledge and understanding required by an individual in leveraging and demonstrating proficiency in working with Advance Automobile Technologies.

Scope

The scope covers the following :

- Perform software updates on various advanced systems.
- Inspecting Advanced Driver Assistance Systems (ADAS) functions.
- Conduct Diagnosis on TPMS, Telematics, Infotainment, Auto HVAC and other advanced features.

Elements and Performance Criteria

Perform software updates on various advanced systems

To be competent, the user/individual on the job must be able to:

PC1. Perform software updates on ECUs to optimize engine performance, fuel efficiency, emissions control, and other parameters.

PC2. Updating ECU software may address issues such as rough idling, hesitation, or improve overall engine operation.

PC3. Install software updates for infotainment systems to improve user interface responsiveness, add new features, fix software bugs, and address security vulnerabilities.

PC4. Perform Software updates for Telematics systems provide remote monitoring, diagnostics, and connectivity services for vehicles.

PC5. Installing updates for telematics may enhance communication protocols, improve data processing algorithms, or add new features such as remote start, vehicle tracking, or over-the-air software updates.

PC6. TPMS software updates would ensure accurate tire pressure readings and timely alerts to drivers.

Inspecting Advanced Driver Assistance Systems (ADAS) functions

To be competent, the user/individual on the job must be able to:

PC7. Visually inspect the vehicle's exterior, looking for any damage to sensors, cameras, or other components associated with ADAS systems. Check for obstructions, dirt, or debris that may interfere with sensor operation.

PC8. Scan the vehicle's onboard computer systems for diagnostic trouble codes related to ADAS components.

PC9. Analyze data from the vehicle's sensors, cameras, and other ADAS components to identify irregularities or discrepancies. Use diagnostic software to review sensor inputs, system outputs, and communication signals for abnormalities.

PC10. Conduct road tests to evaluate the real-world performance of ADAS systems under varying driving conditions. Observe system behavior, responsiveness, and accuracy, noting any anomalies or unexpected behavior.

PC11. Verify the calibration status of sensors such as radar, lidar, cameras, and ultrasonic sensors using calibration targets or alignment tools. Ensure sensors are correctly aligned and calibrated according to manufacturer specifications.

Conduct Diagnosis on TPMS, Telematics, Infotainment, Auto HVAC and other advanced features

To be competent, the user/individual on the job must be able to:

PC12. For TPMS initiate with visually inspecting the tires to check for obvious signs of damage, such as punctures, cuts, or bulges, which could cause tire pressure issues.

PC13. Check the owner's manual or service information to understand the reason behind an illuminated TPMS warning light on the dashboard

PC14. Inspect each tire for the presence of TPMS sensors, which are typically located inside the tire near the valve stem. Check for physical damage, corrosion, or signs of tampering that could affect sensor functionality

PC15. For Telematics begin visually inspects components associated with the telematics system, including antennas, wiring harnesses, and control modules, to check for physical damage, loose connections, or corrosion that could affect system operation

PC16. Performing a reset of the telematics control module or related modules may help resolve communication or functionality issues. Follow manufacturer-recommended procedures for resetting the modules.

PC17. Check the condition and positioning of the vehicle's telematics antennas to ensure they are properly installed and undamaged. Antenna malfunctions can lead to poor signal reception and connectivity issues.

PC18. Use diagnostic equipment to perform network communication tests to ensure that modules within the telematics system can communicate with each other and with the vehicle's main computer system

PC19. For infotainment begin by visually inspecting the infotainment system components, including the display screen, control buttons, knobs, and associated wiring harnesses, for any signs of physical damage or loose connections

PC20. Verify that the infotainment system is receiving power by checking fuses, circuit breakers, and connections to the vehicle's electrical system

PC21. Attempt a system reset if the infotainment system is frozen or unresponsive

PC22. Check for proper operation, responsiveness, and accuracy of each feature by performing functional tests on various features of the infotainment system, including the radio, CD/DVD player, Bluetooth connectivity, navigation system, voice recognition, and touchscreen display

PC23. For HVAC start with visually inspecting the HVAC components, including the compressor, condenser, evaporator, blower motor, ductwork, and refrigerant lines, to check for obvious signs of damage, leaks, or blockages

PC24. Adjust the temperature settings on the HVAC controls and verify whether the air coming from the vents matches the selected temperature. Note any discrepancies for further investigation

PC25. Evaluate the airflow from the vents in various modes to determine if airflow is consistent and sufficient. Check for obstructions in the ductwork or problems with the blower motor

PC26. Connect a manifold gauge set to measure the high and low side pressures of the refrigerant system. Abnormal pressure readings could indicate leaks, blockages, or compressor issues.

PC27. Use a refrigerant leak detector or UV dye, to inspect for refrigerant leaks at connections, fittings, hoses, and components such as the evaporator and condenser. Leaks are repaired as necessary.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Strong understanding of automotive electrical systems is essential for diagnosing and repairing advanced electronic components such as engine control units (ECUs), sensors, actuators, and wiring harnesses

KU2. Familiarity with computer systems and software for troubleshooting electronic control modules, performing software updates, and interpreting diagnostic trouble codes (DTCs) using diagnostic scan tools.

KU3. Need a solid understanding of mechanical systems such as engines, transmissions, drivetrain components, brakes, and suspension systems to diagnose and repair mechanical issues that may impact vehicle performance or safety.

KU4. Knowledge of hybrid and electric vehicle systems, including high-voltage battery packs, electric motors, regenerative braking systems, and onboard charging systems

KU5. Understanding how ADAS components such as radar sensors, cameras, lidar, and ultrasonic sensors work together to provide features like adaptive cruise control, lane-keeping assist, and automatic emergency braking

KU6. Should be familiar with telematics systems and vehicle connectivity features such as remote monitoring, over-the-air software updates, and wireless communication protocols to address issues related to vehicle connectivity and remote diagnostics

KU7. Proficiency in diagnostic techniques such as reading wiring diagrams, using diagnostic scan tools, performing component testing, and interpreting diagnostic trouble codes (DTCs)

KU8. Must adhere to safety procedures and guidelines when working with advanced automobile technologies, especially when dealing with high-voltage systems in hybrid and electric vehicles or performing repairs on ADAS components.

KU9. Engage in continuous learning and professional development to stay updated on the latest technologies, diagnostic techniques, and repair procedures.

KU10. communication and customer service skills are important for interacting with vehicle owners, explaining complex technical issues in a clear and understandable manner, and providing recommendations for vehicle maintenance and repair.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret workplace related documentation

GS2. interpret the needs of customers by understanding the key issues

GS3. communicate using terms, names, grades and other nomenclature pertaining to the automotive trade

GS4. analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently

GS5. identify potential workplace problem and take suitable action

GS6. write in English/regional language

GS7. read various sources of information available for assessing service and repair requirements

GS8. Read policies and regulations pertinent to the job, including OEM guidelines, Health and Safety instructions etc. while working on the Electric Vehicle and its aggregates

GS9. Communicate effectively at the workplace

GS10. Plan work according to the required schedule and location

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Perform software updates on various advanced systems	**5**	**10**	**10**	**-**
PC1. Perform software updates on ECUs to optimize engine performance, fuel efficiency, emissions control, and other parameters.	-	-	-	-
PC2. Updating ECU software may address issues such as rough idling, hesitation, or improve overall engine operation.	-	-	-	-
PC3. Install software updates for infotainment systems to improve user interface responsiveness, add new features, fix software bugs, and address security vulnerabilities.	-	-	-	-
PC4. Perform Software updates for Telematics systems provide remote monitoring, diagnostics, and connectivity services for vehicles.	-	-	-	-
PC5. Installing updates for telematics may enhance communication protocols, improve data processing algorithms, or add new features such as remote start, vehicle tracking, or over-the-air software updates.	-	-	-	-
PC6. TPMS software updates would ensure accurate tire pressure readings and timely alerts to drivers.	-	-	-	-
Inspecting Advanced Driver Assistance Systems (ADAS) functions	**5**	**10**	**10**	**-**
PC7. Visually inspect the vehicle's exterior, looking for any damage to sensors, cameras, or other components associated with ADAS systems. Check for obstructions, dirt, or debris that may interfere with sensor operation.	-	-	-	-
PC8. Scan the vehicle's onboard computer systems for diagnostic trouble codes related to ADAS components.	-	-	-	-
PC9. Analyze data from the vehicle's sensors, cameras, and other ADAS components to identify irregularities or discrepancies. Use diagnostic software to review sensor inputs, system outputs, and communication signals for abnormalities.	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Conduct road tests to evaluate the real-world performance of ADAS systems under varying driving conditions. Observe system behavior, responsiveness, and accuracy, noting any anomalies or unexpected behavior.	-	-	-	-
PC11. Verify the calibration status of sensors such as radar, lidar, cameras, and ultrasonic sensors using calibration targets or alignment tools. Ensure sensors are correctly aligned and calibrated according to manufacturer specifications.	-	-	-	-
Conduct Diagnosis on TPMS, Telematics, Infotainment, Auto HVAC and other advanced features	**10**	**20**	**20**	**-**
PC12. For TPMS initiate with visually inspecting the tires to check for obvious signs of damage, such as punctures, cuts, or bulges, which could cause tire pressure issues.	-	-	-	-
PC13. Check the owner's manual or service information to understand the reason behind an illuminated TPMS warning light on the dashboard	-	-	-	-
PC14. Inspect each tire for the presence of TPMS sensors, which are typically located inside the tire near the valve stem. Check for physical damage, corrosion, or signs of tampering that could affect sensor functionality	-	-	-	-
PC15. For Telematics begin visually inspects components associated with the telematics system, including antennas, wiring harnesses, and control modules, to check for physical damage, loose connections, or corrosion that could affect system operation	-	-	-	-
PC16. Performing a reset of the telematics control module or related modules may help resolve communication or functionality issues. Follow manufacturer-recommended procedures for resetting the modules.	-	-	-	-
PC17. Check the condition and positioning of the vehicle's telematics antennas to ensure they are properly installed and undamaged. Antenna malfunctions can lead to poor signal reception and connectivity issues.	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC18. Use diagnostic equipment to perform network communication tests to ensure that modules within the telematics system can communicate with each other and with the vehicle's main computer system	-	-	-	-
PC19. For infotainment begin by visually inspecting the infotainment system components, including the display screen, control buttons, knobs, and associated wiring harnesses, for any signs of physical damage or loose connections	-	-	-	-
PC20. Verify that the infotainment system is receiving power by checking fuses, circuit breakers, and connections to the vehicle's electrical system	-	-	-	-
PC21. Attempt a system reset if the infotainment system is frozen or unresponsive	-	-	-	-
PC22. Check for proper operation, responsiveness, and accuracy of each feature by performing functional tests on various features of the infotainment system, including the radio, CD/DVD player, Bluetooth connectivity, navigation system, voice recognition, and touchscreen display	-	-	-	-
PC23. For HVAC start with visually inspecting the HVAC components, including the compressor, condenser, evaporator, blower motor, ductwork, and refrigerant lines, to check for obvious signs of damage, leaks, or blockages	-	-	-	-
PC24. Adjust the temperature settings on the HVAC controls and verify whether the air coming from the vents matches the selected temperature. Note any discrepancies for further investigation	-	-	-	-
PC25. Evaluate the airflow from the vents in various modes to determine if airflow is consistent and sufficient. Check for obstructions in the ductwork or problems with the blower motor	-	-	-	-
PC26. Connect a manifold gauge set to measure the high and low side pressures of the refrigerant system. Abnormal pressure readings could indicate leaks, blockages, or compressor issues.	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. Use a refrigerant leak detector or UV dye, to inspect for refrigerant leaks at connections, fittings, hoses, and components such as the evaporator and condenser. Leaks are repaired as necessary.	-	-	-	-
NOS Total	**20**	**40**	**40**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1489
NOS Name	Advance Automobile Technology
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	3.5
Credits	1.5
Version	1.0
Last Reviewed Date	15/03/2023
Next Review Date	15/03/2026
NSQC Clearance Date	15/03/2023

ASC/N9835: Applied Mathematics

Description

This NOS is about to Analyze and optimize various processes, such as cutting speeds, feeds, and tool path planning using Applied Mathematics.

Scope

The scope covers the following :

- Developing mathematical models to represent the processes.
- Applying optimization techniques to find the best possible solutions for process parameters.
- Developing control systems and monitoring strategies to maintain optimal process conditions.

Elements and Performance Criteria

Developing mathematical models to represent the processes

To be competent, the user/individual on the job must be able to:

PC1. Identify the specific process or problem to be modeled, such as optimizing cutting speeds and feeds for a CNC milling operation or planning efficient tool paths for a complex part.

PC2. Gather relevant data and information about the process, including material properties, tool specifications, and machine capabilities.

PC3. Select appropriate mathematical models, equations, or algorithms to represent the process.

PC4. Develop the mathematical model by incorporating the collected data and chosen mathematical formulations.

Applying optimization techniques to find the best possible solutions for process parameters

To be competent, the user/individual on the job must be able to:

PC5. Identify the constraints, like tool life, material properties, or machine capabilities

PC6. Develop a mathematical model that represents the relationship between the process parameters and the desired outcomes.

PC7. Adjust the optimization algorithm's parameters to achieve the best performance for the specific problem.

PC8. Analyze the results obtained from the optimization process, including the optimal values of cutting speeds, feeds, and tool path planning parameters

Developing control systems and monitoring strategies to maintain optimal process conditions.

To be competent, the user/individual on the job must be able to:

PC9. Create mathematical models that represent the dynamic behavior of the process or system under consideration

PC10. Estimate the parameters of the model based on available data from the process, using techniques such as regression analysis, least squares, or maximum likelihood estimation.

PC11. Develop control strategies to regulate the process variables and maintain them within desired operating conditions.

PC12. Apply optimization techniques, such as linear programming, dynamic programming, or genetic algorithms, to minimize costs, maximize throughput, or improve overall process efficiency.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. fundamentals of the CNC/conventional machine
KU2. various types of machining processes such as drilling, boring, turning etc.
KU3. SOP recommended by the manufacturer for using tools, jigs, fixtures, measuring instruments etc., during the machining processes
KU4. how to select and modify the CNC machining program
KU5. SOP recommended by the organisation for operating CNC and conventional machine
KU6. To analyze and optimize various processes, such as cutting speeds, feeds, and tool path planning, using applied mathematics, the following knowledge and understanding skills are essential
KU7. Calculus: Proficiency in differentiation and integration techniques to model and analyze the relationships between process variables, such as cutting speed, feed rate, and tool life
KU8. Differential Equations: Ability to develop and solve differential equations to describe the dynamic behavior of processes and systems, including ordinary and partial differential equations
KU9. Linear Algebra: Understanding of vector and matrix operations, eigenvalues, and eigenvectors to analyze and optimize processes involving multiple variables and constraints
KU10. Optimization Techniques: Knowledge of various optimization methods, such as gradient descent, genetic algorithms, or simulated annealing, to find the optimal values of cutting speed, feed rate, and tool path planning
KU11. Statistics and Probability: Ability to analyze and interpret data collected from the processes, using statistical methods like hypothesis testing, regression analysis, and probability distributions to make informed decisions.
KU12. Discrete Mathematics: Familiarity with combinatorics, graph theory, and network flow analysis to optimize tool path planning and minimize machining time.
KU13. Numerical Methods: Understanding of numerical techniques, such as finite difference, finite element, or boundary element methods, to solve complex problems involving continuous processes.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret work instructions, machine drawings, reports and process documents
GS2. communicate the machining requirements to the seniors and other departments
GS3. communicate issues to the supervisor that occur during machining process
GS4. attentively listen and comprehend the information given by the master technician/team members

GS5. write reports related to production process in English/regional language

GS6. recognise a workplace problem and take suitable action

GS7. analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently

GS8. plan and organise work according to the work requirements

GS9. report to the supervisor or deal with a colleague individually, depending on the type of concern

GS10. complete the assigned tasks with minimum supervision

GS11. suggest improvements (if any) in current ways of working

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Developing mathematical models to represent the processes	**5**	**10**	**10**	**-**
PC1. Identify the specific process or problem to be modeled, such as optimizing cutting speeds and feeds for a CNC milling operation or planning efficient tool paths for a complex part.	1	2	2	-
PC2. Gather relevant data and information about the process, including material properties, tool specifications, and machine capabilities.	2	3	3	-
PC3. Select appropriate mathematical models, equations, or algorithms to represent the process.	1	3	3	-
PC4. Develop the mathematical model by incorporating the collected data and chosen mathematical formulations.	1	2	2	-
Applying optimization techniques to find the best possible solutions for process parameters	**5**	**10**	**10**	**-**
PC5. Identify the constraints, like tool life, material properties, or machine capabilities	1	3	3	-
PC6. Develop a mathematical model that represents the relationship between the process parameters and the desired outcomes.	1	2	2	-
PC7. Adjust the optimization algorithm's parameters to achieve the best performance for the specific problem.	1	3	3	-
PC8. Analyze the results obtained from the optimization process, including the optimal values of cutting speeds, feeds, and tool path planning parameters	2	2	2	-
Developing control systems and monitoring strategies to maintain optimal process conditions.	**5**	**10**	**10**	**-**
PC9. Create mathematical models that represent the dynamic behavior of the process or system under consideration	1	2	2	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Estimate the parameters of the model based on available data from the process, using techniques such as regression analysis, least squares, or maximum likelihood estimation.	1	2	2	-
PC11. Develop control strategies to regulate the process variables and maintain them within desired operating conditions.	1	3	3	-
PC12. Apply optimization techniques, such as linear programming, dynamic programming, or genetic algorithms, to minimize costs, maximize throughput, or improve overall process efficiency.	2	3	3	-
NOS Total	**15**	**30**	**30**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9835
NOS Name	Applied Mathematics
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Generic
NSQF Level	3.5
Credits	3
Version	1.0
Last Reviewed Date	15/03/2023
Next Review Date	15/03/2026
NSQC Clearance Date	15/03/2023

DGT/VSQ/N0104: Employability Skills (120 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the current job market requirement and future of work

PC2. identify and explore learning and employability relevant portals

PC3. research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. for personal growth and the nation's progress

PC5. follow personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC6. follow and promote environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC7. recognize the significance of 21st Century Skills for employment

PC8. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

PC9. adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC10. use English as a medium of formal and informal communication while dealing with topics of everyday conversation in different contexts

PC11. speak over the phone in English, in an audible manner, using appropriate greetings, opening, and closing statements both on personal and work front

PC12. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC13. write short messages, notes, letters, e-mails etc., using accurate English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC14. identify career goals based on the skills, interests, knowledge, and personal attributes

PC15. prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

PC16. follow verbal and non-verbal communication etiquette while communicating in professional and public settings

PC17. use active listening techniques for effective communication

PC18. communicate in writing using appropriate style and format based on formal or informal requirements

PC19. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC20. • ensure personal behaviour, conduct, and use appropriate communication by taking gender into • consideration

PC21. empathize with a PwD and aid a PwD, if asked

PC22. escalate any issues related to sexual harassment at the workplace in accordance with the POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC23. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.

PC24. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook

PC25. identify common components of salary and compute income, expenses, taxes, investments etc

PC26. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC27. operate digital devices and use their features and applications securely and safely

PC28. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.

PC29. display responsible online behaviour while using various social media platforms

PC30. create a personal email account, send and process received messages as per requirement

PC31. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications

PC32. utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC33. identify different types of Entrepreneurship and Enterprises

PC34. use research and networking skills to identify and assess opportunities for potential business

PC35. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC36. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC37. identify different types of customers

PC38. identify and respond to customer requests and needs in a professional manner

PC39. use appropriate tools to collect customer feedback

PC40. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC41. create a professional Curriculum vitae (Résumé)

PC42. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC43. apply to identified job openings using offline /online methods as per requirement

PC44. answer questions politely, with clarity and confidence, during recruitment and selection

PC45. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. different learning and employability related portals

KU3. various constitutional and personal values

KU4. different environmentally sustainable practices and their importance

KU5. Twenty first (21st) century skills and their importance

KU6. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
KU7. importance of career development and setting long- and short-term goals
KU8. Do's and don'ts of effective communication
KU9. POSH Act
KU10. inclusivity and its importance
KU11. different types of disabilities and appropriate verbal and non-verbal communication and behaviour towards PwD
KU12. different types of financial institutes, products, and services
KU13. components of salary and how to compute income and expenditure
KU14. importance of maintaining safety and security in offline and online financial transactions
KU15. different legal rights and laws
KU16. different types of digital devices and the procedure to operate them safely and securely
KU17. how to create and operate an e- mail account
KU18. use applications such as word processors, spreadsheets etc.
KU19. different types of Enterprises and ways to identify business opportunities
KU20. types and needs of customers
KU21. how to apply for a job and prepare for an interview
KU22. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence in English and other languages
GS2. communicate effectively using appropriate language in formal and informal settings
GS3. behave politely and appropriately with all to maintain effective work relationship
GS4. how to work in a virtual mode, using various technological platforms
GS5. perform calculations efficiently
GS6. solve problems effectively
GS7. pay attention to details
GS8. manage time efficiently
GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
Constitutional values – Citizenship	**2**	**1**	**-**	**-**
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. for personal growth and the nation's progress	-	-	-	-
PC5. follow personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC6. follow and promote environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**2**	**3**	**-**	**-**
PC7. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC8. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC9. adopt a continuous learning mindset for personal and professional development	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. use English as a medium of formal and informal communication while dealing with topics of everyday conversation in different contexts	-	-	-	-
PC11. speak over the phone in English, in an audible manner, using appropriate greetings, opening, and closing statements both on personal and work front	-	-	-	-
PC12. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC13. write short messages, notes, letters, e-mails etc., using accurate English	-	-	-	-
Career Development & Goal Setting	**1**	**2**	**-**	**-**
PC14. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC15. prepare a career development plan with short- and long-term goals	-	-	-	-
Communication Skills	**2**	**3**	**-**	**-**
PC16. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC17. use active listening techniques for effective communication	-	-	-	-
PC18. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC19. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**2**	**-**	**-**
PC20. • ensure personal behaviour, conduct, and use appropriate communication by taking gender into • consideration	-	-	-	-
PC21. empathize with a PwD and aid a PwD, if asked	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. escalate any issues related to sexual harassment at the workplace in accordance with the POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC23. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC24. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-
PC25. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC26. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**2**	**3**	**-**	**-**
PC27. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC28. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC29. display responsible online behaviour while using various social media platforms	-	-	-	-
PC30. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC31. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC32. utilize virtual collaboration tools to work effectively	-	-	-	-
Entrepreneurship	**2**	**3**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC33. identify different types of Entrepreneurship and Enterprises	-	-	-	-
PC34. use research and networking skills to identify and assess opportunities for potential business	-	-	-	-
PC35. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC36. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC37. identify different types of customers	-	-	-	-
PC38. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC39. use appropriate tools to collect customer feedback	-	-	-	-
PC40. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**4**	**-**	**-**
PC41. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC42. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC43. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC44. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC45. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0104
NOS Name	Employability Skills (120 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	6
Credits	4
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

ASC/N1492: Workshop Technology Two-Wheeler (OJT)

Description

On the Job Training

Scope

The scope covers the following :

- On the Job Training - OJT

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. On the Job Training - OJT

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. On the Job Training - OJT

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. On the Job Training - OJT

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**-**	**50**	**50**
PC1. On the Job Training - OJT	-	-	50	50
NOS Total	**-**	**-**	**50**	**50**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1492
NOS Name	Workshop Technology Two-Wheeler (OJT)
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	3.5
Credits	11
Version	1.0
Last Reviewed Date	15/03/2023
Next Review Date	15/03/2026
NSQC Clearance Date	15/03/2023

ASC/N1485: Two-Wheeler Technology

Description

This unit describes the knowledge and understanding required by an individual about the technologies employed in the two-wheeler sector to carry out service, repair and overhauling of a two wheeled vehicle.

Scope

The scope covers the following :

- Implementing two-wheeler technologies to practice.
- Prepare to carry out routine service and repairs of two-wheelers.
- Perform post service/repair routine.

Elements and Performance Criteria

Implementing two-wheeler technologies to practice

To be competent, the user/individual on the job must be able to:

PC1. The basics of internal combustion engines, including two-stroke and four-stroke engines, as well as common engine configurations and components need to be learnt and brought to practice.

PC2. Inspecting fuel injection systems, carburetors, ignition systems, and engine management systems to diagnose and troubleshoot engine-related issues.

PC3. Testing different braking systems, including drum brakes, disc brakes, and anti-lock braking systems (ABS).

PC4. Conducting research on brake components such as brake pads, brake discs, brake lines, and master cylinders, and knowing how to inspect, adjust, and replace them.

PC5. Observe the working of suspension components, including front forks, rear shocks, swingarms, and linkages

PC6. Calibrating suspension adjustments such as preload, compression damping, and rebound damping, and observe how they affect ride quality and handling

Prepare to carry out routine service and repairs of two-wheelers

To be competent, the user/individual on the job must be able to:

PC7. Conduct an initial visual inspection of the two-wheeler to assess its overall condition and identify any visible issues or areas of concern.

PC8. Check for damage to bodywork, lights, mirrors, tires, brakes, suspension components, and fluid leaks.

PC9. Communicate with the customer to understand any specific concerns or issues they may have with the vehicle

PC10. Discuss the vehicle's service history, previous repairs, and any recent modifications or upgrades made by the customer.

PC11. Perform pre-service checks, such as checking tire pressures, fluid levels (engine oil, coolant, brake fluid), and battery condition.

PC12. Perform diagnostic tests using scan tools or diagnostic equipment to identify the root cause of the problem

PC13. Maintain the documentation related to inspection, servicing and repair of the vehicle

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The automotive Industry, workshop structure and role and responsibilities of different people in the workshop.

KU2. Standard Operating Procedures (SOP) for receiving vehicles, opening job cards, allocation of work, invoicing, vehicle delivery, handling complaints, etc.

KU3. Different components aggregates as well as auto component manufacturers specifications for the same.

KU4. Fundamental terms and technologies associated with a two-wheeler such as: engine technology, braking systems, suspension systems, electrical systems, fuel systems, knowledge of fuel delivery systems

KU5. Knowledge of tire technology.

KU6. Knowledge Chassis and frame: Understanding chassis geometry, frame types (e.g., tubular, perimeter, monocoque), and frame construction materials (e.g., steel, aluminum, carbon fiber) and interconnection of systems with each other and effect of one system on another

KU7. SOP recommended by OEM for using tools and /equipment for diagnosis or troubleshooting related to electrical/electronic systems such as special service tools, measuring instrument, volt meters, ammeters, ohmmeters, battery tester, dedicated and computer based diagnostic equipment, etc

KU8. Various sources of information available for assessing service and repair requirements of the vehicle including diagnostic displays, visual inspections, test drives, vehicle/equipment manufacturer specifications, tolerance limits of components and options for repair or replacement

KU9. SOP for service, repair and overhauling of electrical/electronic aggregates of the vehicle as prescribed by the OEM.

KU10. Various methods to remove, dismantle, clean, adjust, reassemble and test electrical/electronic components.

KU11. Type and quality of consumables/materials used for the job such as seals, sealant, fasteners, lubricants etc

KU12. How to work on the HV systems which do not require isolation, troubleshooting and replacing parts on the active HV system.

KU13. Safety precautions for equipment and components prescribed by the OEM.

KU14. Specific precautions to ensure that no damage is caused to the vehicle, including its electrical and mechanical aggregates, while carrying out the work

KU15. Organizational/professional code of ethics and standards of practice.

KU16. Documentation requirements for each procedure carried out as part of roles and responsibilities as specified by OEM/auto component manufacturer.

KU17. Various methods to dispose-off replaced failed components/parts in accordance with safety, health and environmental policies and regulation.

KU18. safety, health and environmental policies and regulations for the workplace as well as for automotive trade in general.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret workplace related documentation

GS2. interpret the needs of customers by understanding the key issues

GS3. communicate using terms, names, grades and other nomenclature pertaining to the automotive trade

GS4. analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently

GS5. identify potential workplace problem and take suitable action

GS6. write in English/regional language

GS7. read various sources of information available for assessing service and repair requirements.

GS8. Read policies and regulations pertinent to the job, including OEM guidelines, Health and Safety instructions etc. while working on the Electric Vehicle and its aggregates

GS9. Communicate effectively at the workplace.

GS10. Plan work according to the required schedule and location

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Implementing two-wheeler technologies to practice	**7**	**15**	**15**	**-**
PC1. The basics of internal combustion engines, including two-stroke and four-stroke engines, as well as common engine configurations and components need to be learnt and brought to practice.	-	-	-	-
PC2. Inspecting fuel injection systems, carburetors, ignition systems, and engine management systems to diagnose and troubleshoot engine-related issues.	-	-	-	-
PC3. Testing different braking systems, including drum brakes, disc brakes, and anti-lock braking systems (ABS).	-	-	-	-
PC4. Conducting research on brake components such as brake pads, brake discs, brake lines, and master cylinders, and knowing how to inspect, adjust, and replace them.	-	-	-	-
PC5. Observe the working of suspension components, including front forks, rear shocks, swingarms, and linkages	-	-	-	-
PC6. Calibrating suspension adjustments such as preload, compression damping, and rebound damping, and observe how they affect ride quality and handling	-	-	-	-
Prepare to carry out routine service and repairs of two-wheelers	**8**	**15**	**15**	**-**
PC7. Conduct an initial visual inspection of the two-wheeler to assess its overall condition and identify any visible issues or areas of concern.	-	-	-	-
PC8. Check for damage to bodywork, lights, mirrors, tires, brakes, suspension components, and fluid leaks.	-	-	-	-
PC9. Communicate with the customer to understand any specific concerns or issues they may have with the vehicle	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Discuss the vehicle's service history, previous repairs, and any recent modifications or upgrades made by the customer.	-	-	-	-
PC11. Perform pre-service checks, such as checking tire pressures, fluid levels (engine oil, coolant, brake fluid), and battery condition.	-	-	-	-
PC12. Perform diagnostic tests using scan tools or diagnostic equipment to identify the root cause of the problem	-	-	-	-
PC13. Maintain the documentation related to inspection, servicing and repair of the vehicle	-	-	-	-
NOS Total	**15**	**30**	**30**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1485
NOS Name	Two-Wheeler Technology
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	3.5
Credits	3
Version	1.0
Last Reviewed Date	15/03/2023
Next Review Date	15/03/2026
NSQC Clearance Date	15/03/2023

ASC/N6458: Engineering Drawing

Description

This NOS unit is about reading and interpreting all concepts, symbols, methods, views, etc. of engineering drawing.

Scope

The scope covers the following :

- Interpret information from various views, projection, 2D and 3D shapes Identify drawing standards and symbols
- Modification and storage of drawing

Elements and Performance Criteria

Interpret information from various views, projection, 2D and 3D shapes

To be competent, the user/individual on the job must be able to:

PC1. interpret engineering drawing's uniqueness, dimensions and important features in 2D and 3D shapes

PC2. identify the difference between 2D and 3D shapes

PC3. explain difference between first angle projection and third angle projection in mechanical engineering drawing

PC4. interpret all the 3 axes (x, y and z axis) and geometrical shapes (cones, cylinder, sphere, cuboid, etc) on to a 2D and 3D projection

PC5. identify details of the machine component which are not clearly visible by interpreting section views

Identify drawing standards and symbols

To be competent, the user/individual on the job must be able to:

PC6. interpret Geometric Dimensioning and Tolerencing (GD&T) symbols in the drawings

PC7. interpret symbols of Radius, controlled radius, spherical radius, diameter, spherical diameter, square, counterbore, spotface, depth, countersink, “by”, maximum dimension, minimum dimension, reference, dimension origin etc

PC8. identify the sequence of operations which enables the selection and prioritization of the datums

PC9. read and interpret information from Tolerance Zone boundaries for part features in terms of shape and size

Modification and storage of drawing

To be competent, the user/individual on the job must be able to:

PC10. observe any modification, changes required in the drawing and communicate the same to the concerned team in the organization

PC11. store the drawings in an easily accessible place, avoiding damage from moisture, chemicals and fire

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. relevant organisational standards such as work standard, Standard Operating Procedure, quality process, maintenance standards etc. followed in the company
KU2. importance of cycle-time and required output as per work order and work instructions
KU3. drawing standards used by the company
KU4. use of drawing tools such as scales, compass, types of pencils, CAD and CAM software etc.
KU5. the basics of engineering drawing, orthographic projection, isometric projection, GD&T etc.
KU6. importance of various projections, views, symbols and dimensions of drawing
KU7. use of geometric shapes like lines, angles, circles, etc for interpreting the drawing

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret workplace related drawing
GS2. communicate the changes and requirements to supervisor by using relevant drawing terms and nomenclature
GS3. attentively listen and comprehend the information given by the supervisor/team members
GS4. write in English/regional language
GS5. recognise problem in drawing and take suitable action
GS6. analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Interpret information from various views, projection, 2D and 3D shapes	**5**	**10**	**10**	**-**
PC1. interpret engineering drawing's uniqueness, dimensions and important features in 2D and 3D shapes	-	-	-	-
PC2. identify the difference between 2D and 3D shapes	-	-	-	-
PC3. explain difference between first angle projection and third angle projection in mechanical engineering drawing	-	-	-	-
PC4. interpret all the 3 axes (x, y and z axis) and geometrical shapes (cones, cylinder, sphere, cuboid, etc) on to a 2D and 3D projection	-	-	-	-
PC5. identify details of the machine component which are not clearly visible by interpreting section views	-	-	-	-
Identify drawing standards and symbols	**5**	**15**	**15**	**-**
PC6. interpret Geometric Dimensioning and Tolerencing (GD&T) symbols in the drawings	-	-	-	-
PC7. interpret symbols of Radius, controlled radius, spherical radius, diameter, spherical diameter, square, counterbore, spotface, depth, countersink, “by”, maximum dimension, minimum dimension, reference, dimension origin etc	-	-	-	-
PC8. identify the sequence of operations which enables the selection and prioritization of the datums	-	-	-	-
PC9. read and interpret information from Tolerance Zone boundaries for part features in terms of shape and size	-	-	-	-
Modification and storage of drawing	**5**	**5**	**5**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. observe any modification, changes required in the drawing and communicate the same to the concerned team in the organization	-	-	-	-
PC11. store the drawings in an easily accessible place, avoiding damage from moisture, chemicals and fire	-	-	-	-
NOS Total	**15**	**30**	**30**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N6458
NOS Name	Engineering Drawing
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Production Engineering
NSQF Level	3.5
Credits	3
Version	2.0
Last Reviewed Date	15/03/2023
Next Review Date	15/03/2026
NSQC Clearance Date	15/03/2023

ASC/N1490: Electric Two or Three Wheeler Technology

Description

This unit describes how industry professionals can embrace electric two or three-wheeler technologies

Scope

The scope covers the following :

- Working with advanced electric two-wheelers.
- Implementing the knowledge acquired to work with electric three-wheelers.

Elements and Performance Criteria

Perform software updates on various advanced systems

To be competent, the user/individual on the job must be able to:

PC1. Inspect the electric vehicle's battery pack for signs of damage, corrosion, or wear

PC2. Check battery connections, clean terminals, and ensure proper mounting. Perform battery capacity tests to assess overall health and performance.

PC3. Verify the functionality of the electric vehicle charging system, including the charging port, onboard charger, and charging cable.

PC4. Conduct tests for checking for proper voltage output, inspect cable integrity, and troubleshoot any charging-related issues

PC5. Examine the electric motor, drivetrain components, and associated wiring for wear, damage, or abnormalities.

PC6. Perform tests to ensure proper motor operation and diagnose any issues affecting propulsion.

PC7. Use diagnostic tools to assess the performance of the vehicle's motor controller, which regulates power delivery from the battery to the electric motor.

PC8. Check for error codes, perform system tests, and troubleshoot controller-related problems.

PC9. Inspect the electric two-wheeler's braking system, including hydraulic or regenerative braking components.

PC10. Examine the suspension system, including forks, shocks, and swingarm, for wear, leaks, or damage. Also check steering components for proper alignment and functionality to ensure stability and control.

PC11. Inspect tires for wear, damage, and proper inflation. Check tread depth, sidewall condition, and tire pressure and recommend tire replacement or rotation if needed for optimal performance and safety.

Implementing the knowledge acquired to work with electric three-wheelers

To be competent, the user/individual on the job must be able to:

PC12. Assess the condition of the vehicle's battery pack, including checking for physical damage, monitoring state of charge (SOC), and inspecting individual cell voltages

PC13. Inspect the charging system components, including the onboard charger, charging port, and charging cables, to ensure they are functioning properly.

PC14. Inspect the braking components, including brake pads, discs, calipers, and hydraulic lines, for wear and tear. Also check brake fluid levels and perform brake bleeding if necessary to maintain proper braking performance

PC15. Check wheel alignment and inspect tires for signs of wear, damage, or improper inflation. Adjust wheel alignment as needed and replace tires if they are worn or damaged.

PC16. Inspect the electric motor, drivetrain components, and transmission (if applicable) for signs of wear, damage, or malfunction. Lubricate moving parts and replace worn components as necessary to ensure smooth operation.

PC17. Perform a comprehensive inspection of the vehicle's electrical system, including wiring harnesses, connectors, switches, and sensors. Check for loose connections, damaged wires, and signs of corrosion.

PC18. Update the vehicle's software/firmware to the latest version provided by the manufacturer. This may involve recalibrating control systems, updating vehicle settings, or addressing software bugs and glitches.

PC19. Inspect safety systems such as lights, turn signals, horn, and emergency braking systems to ensure they are functioning properly. They replace faulty bulbs or components and test system functionality.

PC20. Examine suspension components, including shocks, struts, springs, and steering linkage, for wear and damage. They lubricate moving parts and replace worn components to maintain ride comfort and handling.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Strong understanding of automotive electrical systems is essential for diagnosing and repairing advanced electronic components such as engine control units (ECUs), sensors, actuators, and wiring harnesses.

KU2. Familiarity with computer systems and software for troubleshooting electronic control modules, performing software updates, and interpreting diagnostic trouble codes (DTCs) using diagnostic scan tools.

KU3. Need a solid understanding of mechanical systems such as engines, transmissions, drivetrain components, brakes, and suspension systems to diagnose and repair mechanical issues that may impact vehicle performance or safety.

KU4. Knowledge of hybrid and electric vehicle systems, including high-voltage battery packs, electric motors, regenerative braking systems, and onboard charging systems

KU5. Understanding how ADAS components such as radar sensors, cameras, lidar, and ultrasonic sensors work together to provide features like adaptive cruise control, lane-keeping assist, and automatic emergency braking.

KU6. Should be familiar with telematics systems and vehicle connectivity features such as remote monitoring, over-the-air software updates, and wireless communication protocols to address issues related to vehicle connectivity and remote diagnostics.

KU7. Proficiency in diagnostic techniques such as reading wiring diagrams, using diagnostic scan tools, performing component testing, and interpreting diagnostic trouble codes (DTCs)

KU8. Must adhere to safety procedures and guidelines when working with advanced automobile technologies, especially when dealing with high-voltage systems in hybrid and electric vehicles or performing repairs on ADAS components.

KU9. Engage in continuous learning and professional development to stay updated on the latest technologies, diagnostic techniques, and repair procedures.

KU10. communication and customer service skills are important for interacting with vehicle owners, explaining complex technical issues in a clear and understandable manner, and providing recommendations for vehicle maintenance and repair.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret workplace related documentation

GS2. interpret the needs of customers by understanding the key issues

GS3. communicate using terms, names, grades and other nomenclature pertaining to the automotive trade

GS4. analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently

GS5. identify potential workplace problem and take suitable action

GS6. write in English/regional language

GS7. read various sources of information available for assessing service and repair requirements.

GS8. Read policies and regulations pertinent to the job, including OEM guidelines, Health and Safety instructions etc. while working on the Electric Vehicle and its aggregates.

GS9. Communicate effectively at the workplace.

GS10. Plan work according to the required schedule and location

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Perform software updates on various advanced systems	**5**	**10**	**10**	**-**
PC1. Inspect the electric vehicle's battery pack for signs of damage, corrosion, or wear	-	-	-	-
PC2. Check battery connections, clean terminals, and ensure proper mounting. Perform battery capacity tests to assess overall health and performance.	-	-	-	-
PC3. Verify the functionality of the electric vehicle charging system, including the charging port, onboard charger, and charging cable.	-	-	-	-
PC4. Conduct tests for checking for proper voltage output, inspect cable integrity, and troubleshoot any charging-related issues	-	-	-	-
PC5. Examine the electric motor, drivetrain components, and associated wiring for wear, damage, or abnormalities.	-	-	-	-
PC6. Perform tests to ensure proper motor operation and diagnose any issues affecting propulsion.	-	-	-	-
PC7. Use diagnostic tools to assess the performance of the vehicle's motor controller, which regulates power delivery from the battery to the electric motor.	-	-	-	-
PC8. Check for error codes, perform system tests, and troubleshoot controller-related problems.	-	-	-	-
PC9. Inspect the electric two-wheeler's braking system, including hydraulic or regenerative braking components.	-	-	-	-
PC10. Examine the suspension system, including forks, shocks, and swingarm, for wear, leaks, or damage. Also check steering components for proper alignment and functionality to ensure stability and control.	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Inspect tires for wear, damage, and proper inflation. Check tread depth, sidewall condition, and tire pressure and recommend tire replacement or rotation if needed for optimal performance and safety.	-	-	-	-
Implementing the knowledge acquired to work with electric three-wheelers	**10**	**20**	**20**	**-**
PC12. Assess the condition of the vehicle's battery pack, including checking for physical damage, monitoring state of charge (SOC), and inspecting individual cell voltages	-	-	-	-
PC13. Inspect the charging system components, including the onboard charger, charging port, and charging cables, to ensure they are functioning properly.	-	-	-	-
PC14. Inspect the braking components, including brake pads, discs, calipers, and hydraulic lines, for wear and tear. Also check brake fluid levels and perform brake bleeding if necessary to maintain proper braking performance	-	-	-	-
PC15. Check wheel alignment and inspect tires for signs of wear, damage, or improper inflation. Adjust wheel alignment as needed and replace tires if they are worn or damaged.	-	-	-	-
PC16. Inspect the electric motor, drivetrain components, and transmission (if applicable) for signs of wear, damage, or malfunction. Lubricate moving parts and replace worn components as necessary to ensure smooth operation.	-	-	-	-
PC17. Perform a comprehensive inspection of the vehicle's electrical system, including wiring harnesses, connectors, switches, and sensors. Check for loose connections, damaged wires, and signs of corrosion.	-	-	-	-
PC18. Update the vehicle's software/firmware to the latest version provided by the manufacturer. This may involve recalibrating control systems, updating vehicle settings, or addressing software bugs and glitches.	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. Inspect safety systems such as lights, turn signals, horn, and emergency braking systems to ensure they are functioning properly. They replace faulty bulbs or components and test system functionality.	-	-	-	-
PC20. Examine suspension components, including shocks, struts, springs, and steering linkage, for wear and damage. They lubricate moving parts and replace worn components to maintain ride comfort and handling.	-	-	-	-
NOS Total	**15**	**30**	**30**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1490
NOS Name	Electric Two or Three Wheeler Technology
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	3.5
Credits	3
Version	1.0
Last Reviewed Date	15/03/2023
Next Review Date	15/03/2026
NSQC Clearance Date	15/03/2023

ASC/N1491: Four-Wheeler Technology

Description

This unit describes the continuous advancements and comprehensive adoption of Four-Wheeler Technologies by industry professionals.

Scope

The scope covers the following :

- Continuous adaptation with changing emission control norms.
- Working with Hybrid Vehicle Systems.
- Conduct Diagnosis on Safety Systems and advanced driver aid systems.

Elements and Performance Criteria

Continuous adaptation with changing emission control norms

To be competent, the user/individual on the job must be able to:

PC1. Compliance with emissions regulations is a critical aspect of vehicle maintenance

PC2. Technicians should understand emission control systems such as catalytic converters, exhaust gas recirculation (EGR) systems, and selective catalytic reduction (SCR) systems

PC3. Diagnostic scan tools should be used to retrieve diagnostic trouble codes (DTCs) stored in the vehicle's onboard computer related to emission control system issues. Interpret the codes to identify the specific components or systems causing emissions-related problems.

PC4. Use exhaust gas analyzers, technicians measure the composition of exhaust gases to ensure they fall within acceptable emissions limits.

PC5. Analyze readings of oxygen ($O2$), carbon dioxide ($CO2$), carbon monoxide (CO), hydrocarbons (HC), and nitrogen oxides (NOx) to diagnose engine performance issues and emissions problems.

PC6. Test oxygen sensors (both upstream and downstream) to ensure they are functioning properly. Check sensor voltages and response times to determine if sensors are accurately detecting oxygen levels in the exhaust gases.

Working with Hybrid Vehicle Systems

To be competent, the user/individual on the job must be able to:

PC7. Inspect the hybrid battery pack to assess its health and performance. This may involve checking the battery's state of charge (SOC), voltage levels, cell balance, and overall capacity using diagnostic tools

PC8. Examine the battery cooling system, including coolant levels, hoses, and components, to ensure proper cooling of the battery pack. Check for leaks, blockages, or malfunctions that could affect battery performance or safety.

PC9. Follow safety protocols and procedures for working with high-voltage components in hybrid vehicles.

PC10. Ensure proper insulation, isolation, and disconnection of high-voltage systems to prevent electric shock hazards during maintenance and repair.

PC11. Perform routine maintenance tasks on the ICE, such as oil changes, filter replacements, and spark plug inspections, to ensure optimal engine performance and efficiency. Also check for leaks, belt tension, and other mechanical issues

PC12. Inspect the electric motor, inverter, and associated components for signs of wear, damage, or malfunction. Also check electrical connections, wiring harnesses, and cooling systems to ensure proper operation of the electric propulsion system.

PC13. Test the regenerative braking system to ensure it is functioning properly. They verify that the system can effectively capture kinetic energy during braking and recharge the hybrid battery pack.

Conduct Diagnosis on Safety Systems and advanced driver aid systems

To be competent, the user/individual on the job must be able to:

PC14. Diagnose the airbag system components, including airbag modules, sensors, and the airbag control module. They check for signs of damage, corrosion, or malfunction that could affect the deployment of airbags during a collision

PC15. Examine seat belts for wear, damage, or fraying. Ensure that seat belt retractors, buckles, and anchors are functioning properly and that seat belts latch securely and retract smoothly.

PC16. Check the ABS components, including sensors, modulator valves, and the ABS control module. Verify that the ABS warning light illuminates during system self-check and that ABS functions operate correctly during braking

PC17. Inspect ESC components, including sensors, actuators, and the ESC control module. They ensure that ESC functions such as traction control and stability control operate correctly, especially during evasive maneuvers or slippery road conditions.

PC18. Verify that sensors such as radar, cameras, lidar, and ultrasonic sensors are properly calibrated. Misalignment or incorrect calibration can lead to inaccurate readings and affect the performance of ADAS features.

PC19. Ensure that sensors are securely mounted and properly aligned according to manufacturer specifications. Then check for physical damage, obstructions, or misalignment that could interfere with sensor functionality.

PC20. Clean sensors to remove dirt, debris, and obstructions that can impair sensor performance. Clean sensors provide accurate data for ADAS features such as adaptive cruise control, lane-keeping assist, and automatic emergency braking

PC21. For HVAC start with visually inspecting the HVAC components, including the compressor, condenser, evaporator, blower motor, ductwork, and refrigerant lines, to check for obvious signs of damage, leaks, or blockages.

PC22. Adjust the temperature settings on the HVAC controls and verify whether the air coming from the vents matches the selected temperature. Note any discrepancies for further.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Should understand the components of an Hybrid powertrain, including the electric motor, motor controller (inverter), battery pack, charger, internal combustion engine and onboard charging system.

KU2. Familiarity with computer systems and software for troubleshooting electronic control modules, performing software updates, and interpreting diagnostic trouble codes (DTCs) using diagnostic scan tools.

KU3. Need a solid understanding of mechanical systems such as engines, transmissions, drivetrain components, brakes, and suspension systems to diagnose and repair mechanical issues that may impact vehicle performance or safety

KU4. Knowledge of hybrid and electric vehicle systems, including high-voltage battery packs, electric motors, regenerative braking systems, and onboard charging systems

KU5. Understanding how ADAS components such as radar sensors, cameras, lidar, and ultrasonic sensors work together to provide features like adaptive cruise control, lane-keeping assist, and automatic emergency braking.

KU6. Proficiency in diagnostic techniques such as reading wiring diagrams, using diagnostic scan tools, performing component testing, and interpreting diagnostic trouble codes (DTCs)

KU7. Should be familiar with local, national, and international emissions regulations, including standards set by organizations such as the Environmental Protection Agency (EPA) in the United States or the European Union's Euro standards. Understanding emission limits and testing procedures is crucial for ensuring vehicles comply with regulatory requirements.

KU8. Understanding of regenerative braking principles and how regenerative braking systems function to recover kinetic energy during braking, improve energy efficiency, and extend driving range

KU9. Need to understand fuel system components such as fuel injectors, fuel pressure regulators, fuel pumps, and fuel filters. They should be able to diagnose issues related to fuel delivery, air-fuel mixture, and fuel system leaks that may impact emissions.

KU10. Should have knowledge of exhaust system components, including mufflers, resonators, catalytic converters, and exhaust pipes. Understanding exhaust gas flow, heat management, and the role of catalytic converters in reducing harmful emissions is essential.

KU11. Must adhere to safety procedures and guidelines when working with advanced automobile technologies, especially when dealing with high-voltage systems in hybrid and electric vehicles or performing repairs on ADAS components

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret workplace related documentation

GS2. interpret the needs of customers by understanding the key issues

GS3. communicate using terms, names, grades and other nomenclature pertaining to the automotive trade

GS4. analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently

GS5. identify potential workplace problem and take suitable action

GS6. write in English/regional language

GS7. read various sources of information available for assessing service and repair requirements

GS8. Read policies and regulations pertinent to the job, including OEM guidelines, Health and Safety instructions etc. while working on the Electric Vehicle and its aggregates

GS9. Communicate effectively at the workplace

GS10. Plan work according to the required schedule and location

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Continuous adaptation with changing emission control norms	**5**	**10**	**10**	**-**
PC1. Compliance with emissions regulations is a critical aspect of vehicle maintenance	-	-	-	-
PC2. Technicians should understand emission control systems such as catalytic converters, exhaust gas recirculation (EGR) systems, and selective catalytic reduction (SCR) systems	-	-	-	-
PC3. Diagnostic scan tools should be used to retrieve diagnostic trouble codes (DTCs) stored in the vehicle's onboard computer related to emission control system issues. Interpret the codes to identify the specific components or systems causing emissions-related problems.	-	-	-	-
PC4. Use exhaust gas analyzers, technicians measure the composition of exhaust gases to ensure they fall within acceptable emissions limits.	-	-	-	-
PC5. Analyze readings of oxygen (O2), carbon dioxide (CO2), carbon monoxide (CO), hydrocarbons (HC), and nitrogen oxides (NOx) to diagnose engine performance issues and emissions problems.	-	-	-	-
PC6. Test oxygen sensors (both upstream and downstream) to ensure they are functioning properly. Check sensor voltages and response times to determine if sensors are accurately detecting oxygen levels in the exhaust gases.	-	-	-	-
Working with Hybrid Vehicle Systems	**5**	**10**	**10**	**-**
PC7. Inspect the hybrid battery pack to assess its health and performance. This may involve checking the battery's state of charge (SOC), voltage levels, cell balance, and overall capacity using diagnostic tools	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. Examine the battery cooling system, including coolant levels, hoses, and components, to ensure proper cooling of the battery pack. Check for leaks, blockages, or malfunctions that could affect battery performance or safety.	-	-	-	-
PC9. Follow safety protocols and procedures for working with high-voltage components in hybrid vehicles.	-	-	-	-
PC10. Ensure proper insulation, isolation, and disconnection of high-voltage systems to prevent electric shock hazards during maintenance and repair.	-	-	-	-
PC11. Perform routine maintenance tasks on the ICE, such as oil changes, filter replacements, and spark plug inspections, to ensure optimal engine performance and efficiency. Also check for leaks, belt tension, and other mechanical issues	-	-	-	-
PC12. Inspect the electric motor, inverter, and associated components for signs of wear, damage, or malfunction. Also check electrical connections, wiring harnesses, and cooling systems to ensure proper operation of the electric propulsion system.	-	-	-	-
PC13. Test the regenerative braking system to ensure it is functioning properly. They verify that the system can effectively capture kinetic energy during braking and recharge the hybrid battery pack.	-	-	-	-
Conduct Diagnosis on Safety Systems and advanced driver aid systems	**5**	**10**	**10**	**-**
PC14. Diagnose the airbag system components, including airbag modules, sensors, and the airbag control module. They check for signs of damage, corrosion, or malfunction that could affect the deployment of airbags during a collision	-	-	-	-
PC15. Examine seat belts for wear, damage, or fraying. Ensure that seat belt retractors, buckles, and anchors are functioning properly and that seat belts latch securely and retract smoothly.	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. Check the ABS components, including sensors, modulator valves, and the ABS control module. Verify that the ABS warning light illuminates during system self-check and that ABS functions operate correctly during braking	-	-	-	-
PC17. Inspect ESC components, including sensors, actuators, and the ESC control module. They ensure that ESC functions such as traction control and stability control operate correctly, especially during evasive maneuvers or slippery road conditions.	-	-	-	-
PC18. Verify that sensors such as radar, cameras, lidar, and ultrasonic sensors are properly calibrated. Misalignment or incorrect calibration can lead to inaccurate readings and affect the performance of ADAS features.	-	-	-	-
PC19. Ensure that sensors are securely mounted and properly aligned according to manufacturer specifications. Then check for physical damage, obstructions, or misalignment that could interfere with sensor functionality.	-	-	-	-
PC20. Clean sensors to remove dirt, debris, and obstructions that can impair sensor performance. Clean sensors provide accurate data for ADAS features such as adaptive cruise control, lane-keeping assist, and automatic emergency braking	-	-	-	-
PC21. For HVAC start with visually inspecting the HVAC components, including the compressor, condenser, evaporator, blower motor, ductwork, and refrigerant lines, to check for obvious signs of damage, leaks, or blockages.	-	-	-	-
PC22. Adjust the temperature settings on the HVAC controls and verify whether the air coming from the vents matches the selected temperature. Note any discrepancies for further.	-	-	-	-
NOS Total	**15**	**30**	**30**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1491
NOS Name	Four-Wheeler Technology
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	3.5
Credits	3
Version	1.0
Last Reviewed Date	15/03/2023
Next Review Date	15/03/2026
NSQC Clearance Date	15/03/2023

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Components of Assessment: - Each subject will be assessed in three components: Theory (20% weightage), Practical (40% weightage), and On-job Training (OJT, 40% weightage).

2. Passing Parameters: - To pass the semester, students must meet both the assessment parameters given below.

Parameter 1 - Weighted Semester Score: - Students must achieve a minimum of 60% in the weighted average score across all three components (Theory, Practical, and OJT) for each subject.

Parameter 2 - Individual Component Score: - Students need to score at least 40% in each individual component (Theory, Practical, and OJT) of every subject.

Mandatory Note: This qualification can be offered as part of a Diploma program, in line with the 39th NSQC, ASDC Diploma (Diploma in Manufacturing Technology) approval. However, achieving 40 credits in a year is mandatory for progression within the Diploma course. Therefore, it is required to select at least one optional NOS in every semester to meet this requirement.

Minimum Aggregate Passing % at QP Level : 40

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N6314.Metrology (Measurement)	20	40	40	-	100	10
ASC/N1483.Basic of Automobile Technology	20	40	40	0	100	10
ASC/N1484.Tools & Equipment	20	40	40	0	100	10
ASC/N1486.Workshop Safety	15	30	30	0	75	10
ASC/N1487.Automobile Electrical System	20	40	40	0	100	10
ASC/N1488.Automobile Electronics	20	40	40	0	100	10
ASC/N1489.Advance Automobile Technology	20	40	40	0	100	10
ASC/N9835.Applied Mathematics	15	30	30	-	75	10
DGT/VSQ/N0104.Employability Skills (120 Hours)	20	30	-	-	50	10
ASC/N1492.Workshop Technology Two-Wheeler (OJT)	0	0	50	50	100	10
Total	**170**	**330**	**350**	**50**	**900**	**100**

Optional: 1 Semester 1: Two-Wheeler Technology

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N1485.Two-Wheeler Technology	15	30	30	0	75	50
Total	**15**	**30**	**30**	**-**	**75**	**50**

Optional: 2 Semester 1: Engineering Drawing

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N6458.Engineering Drawing	15	30	30	0	75	50
Total	**15**	**30**	**30**	**-**	**75**	**50**

Optional: 3 Semester 2: Electric Two or Three Wheeler Technology

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N1490.Electric Two or Three Wheeler Technology	15	30	30	0	75	50
Total	**15**	**30**	**30**	**-**	**75**	**50**

Optional: 4 Semester 2: Four-Wheeler Technology

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N1491.Four-Wheeler Technology	15	30	30	0	75	50
Total	**15**	**30**	**30**	**-**	**75**	**50**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.